

Institute of Computer Engineering Technology



iCET Certified Professional

Coursework

Module	Object Oriented Programming
Name	Student Management System

Case Study

The Institute of Computer Engineering Technology (iCET) is a software engineering training institute in Sri Lanka, with over 1,000 students actively engaged in learning. Therefore, the iCET management has decided to upgrade its student management system to better handle student registration and monitor academic progress. The following use case outlines the system's scope and functionalities.

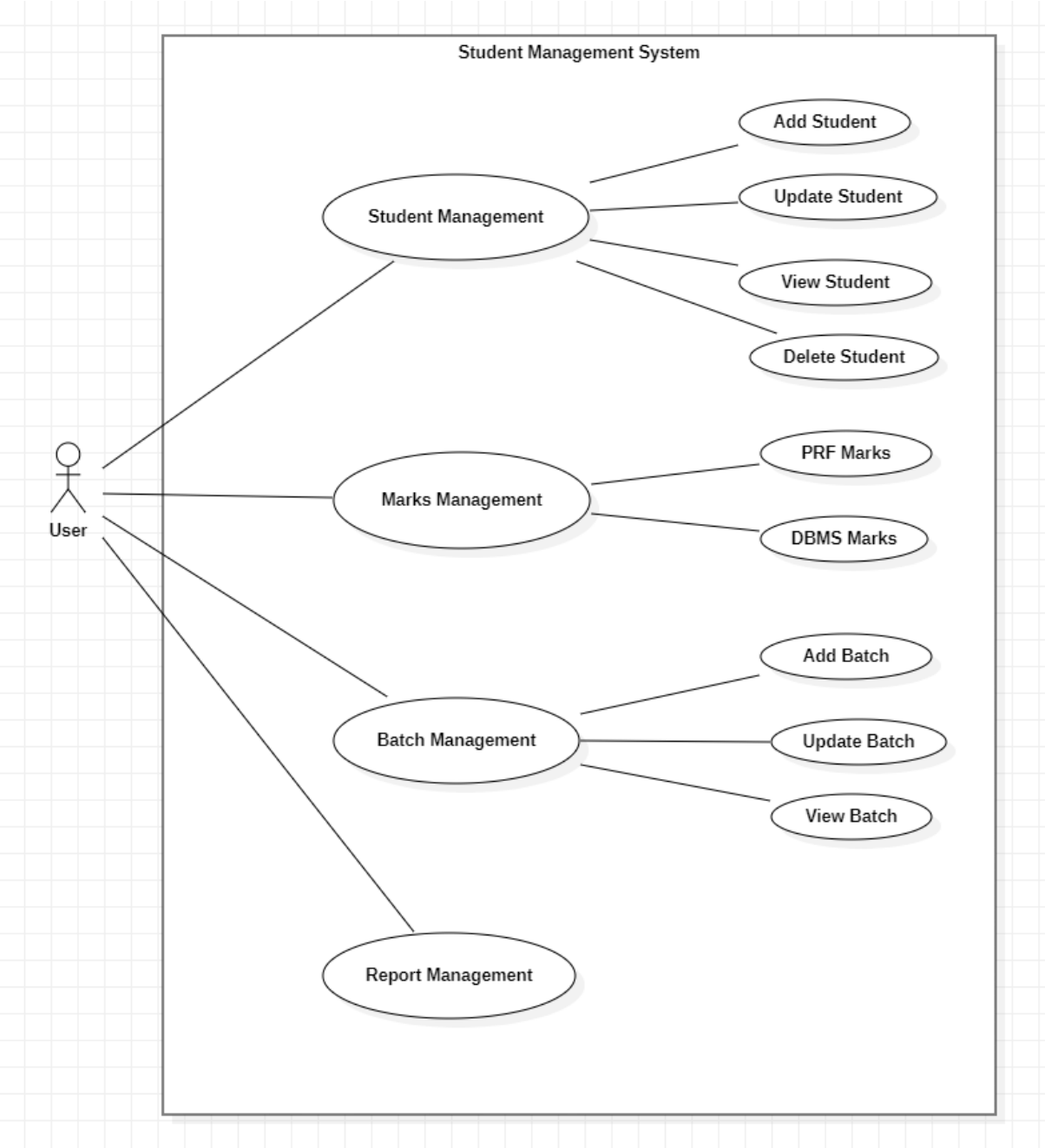


Figure 1 - iCET Student Management System UseCase Diagram

Requirements

You are required to create a Java Swing GUI application for a Student Management System. In this application, you need to implement the following use cases.

When the application is executed, it should display a Graphical User Interface (GUI) similar to the example below, allowing the user to click on the corresponding option they wish to execute. This will serve as the Home Page of the application you are developing.



Figure 2 - Home Page of the iCET Student Management System

1. Student Management

There are four options in Student Management, and the user can click the corresponding option they wish to execute. Below is the Student Management interface you should develop.

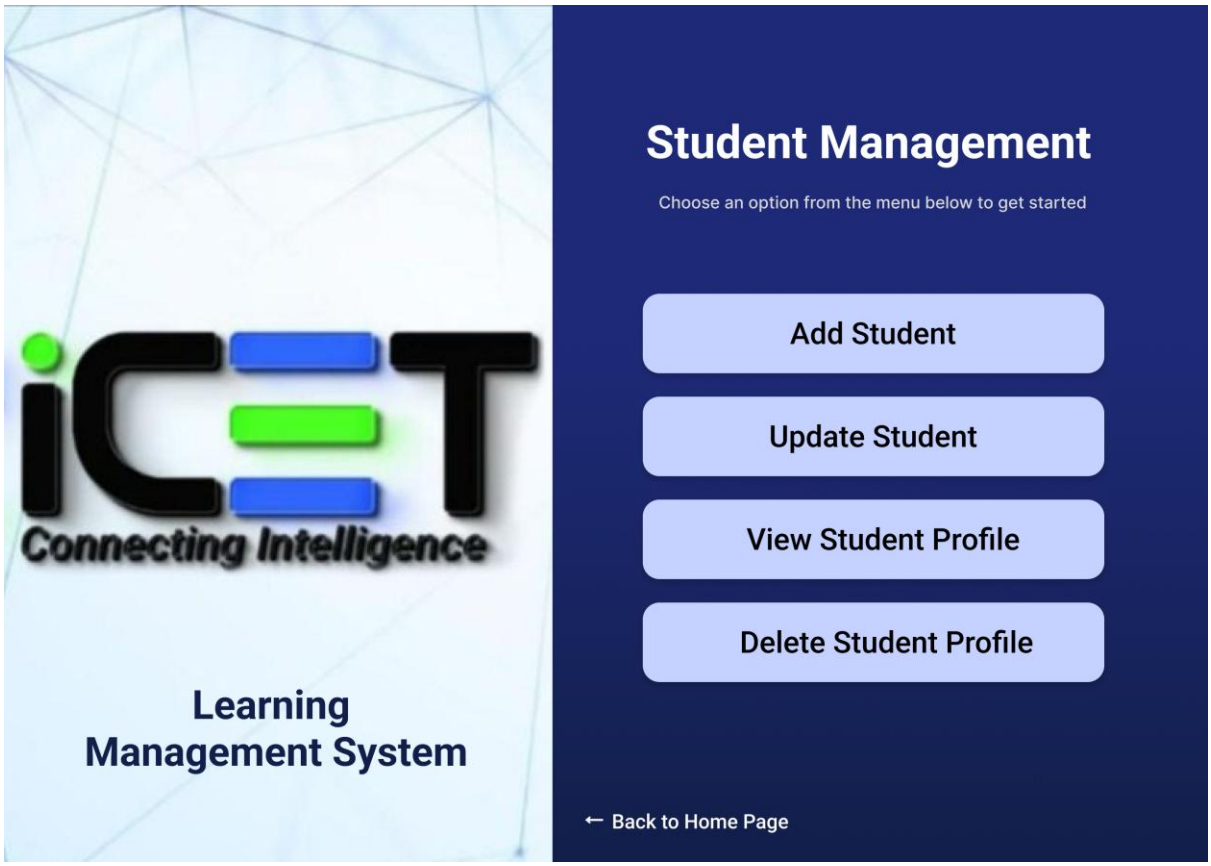


Figure 3- Student Management Home Page

1.1 Add Student

When adding a new student, the user must first select the batch from the **dropdown** to which the student will be added.

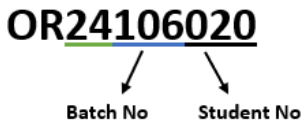
Batch Details:

- Batch Name – Each batch has a unique number starting from 100.
- Batch Status – There are two possible statuses for a batch:
    - ENROLLMENTOPEN = 1
    - ENROLLMENTCLOSED = 0

If the selected batch status is ENROLLMENTCLOSED, the system should prevent the user from adding students to that batch and inform the user accordingly. If the batch status is ENROLLMENTOPEN, the user can proceed to add a new student to the system. The system stores five key details for each student:

1. ***Student ID*** – This ID is generated automatically by the system and consists of 10 characters. If the student is registered for physical lectures, the ID should start with 'PR'. If the student is registered for online lectures, the ID should start with 'OR'.

The remaining characters must follow a specific format and be sequential, based on the last Student ID in the selected batch. Student IDs must be unique and cannot be generated randomly.



*Every student ID should start with '24'.
*Online Lectures -'OR' Physical Lectures – 'PR'.

2. ***Student NIC*** – The user must enter the student's NIC. The system uses the NIC to check whether the student already exists in the system. If the NIC is found, the system should display the existing student profile. If the NIC is not found, the user should enter the rest of the student details before adding the student.

3. ***Student Name*** – Student names can be duplicated.

4. ***PRF Marks*** – When adding a new student, the default marks should be set to **-2**, which indicates that the exam has not been conducted.

5. ***DBMS Marks*** – When adding a new student, the default marks should be set to **-2**, which indicates that the exam has not been conducted.

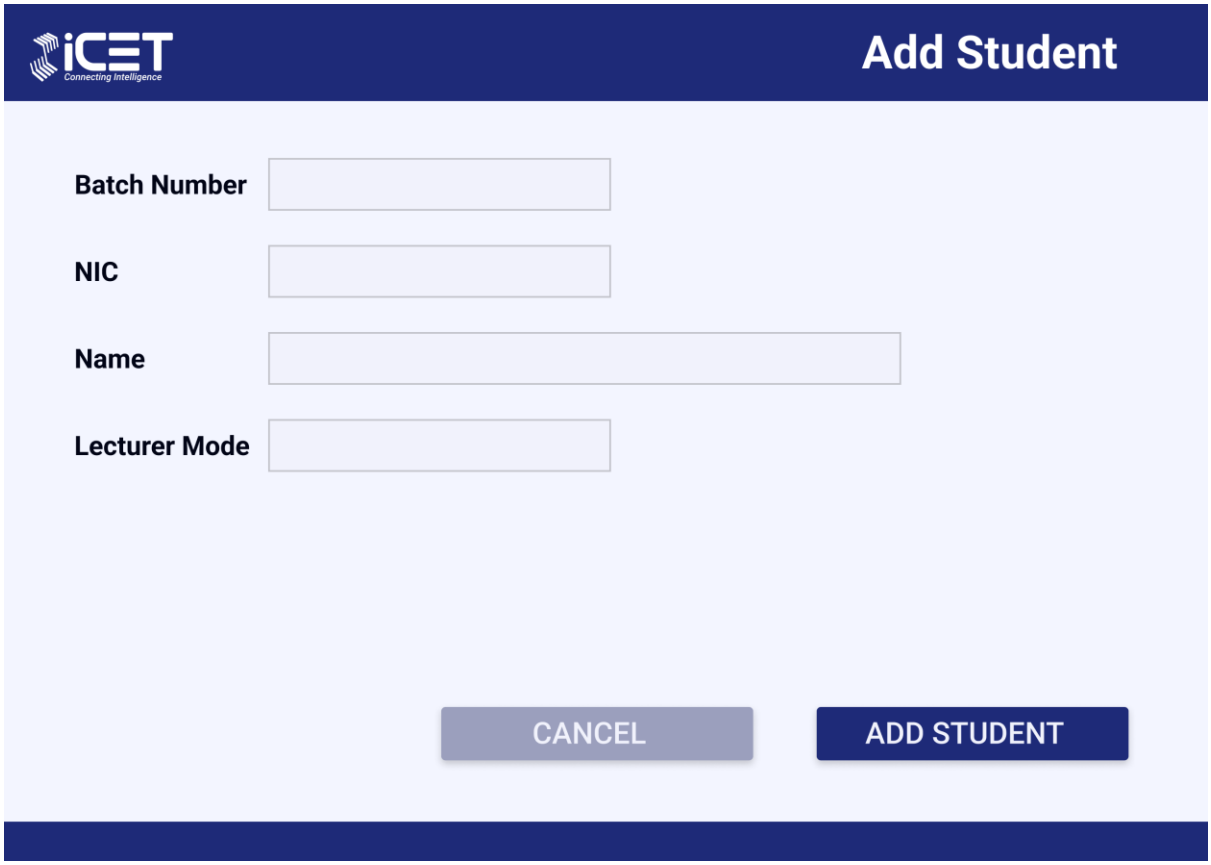


Figure 4 - Add New Student

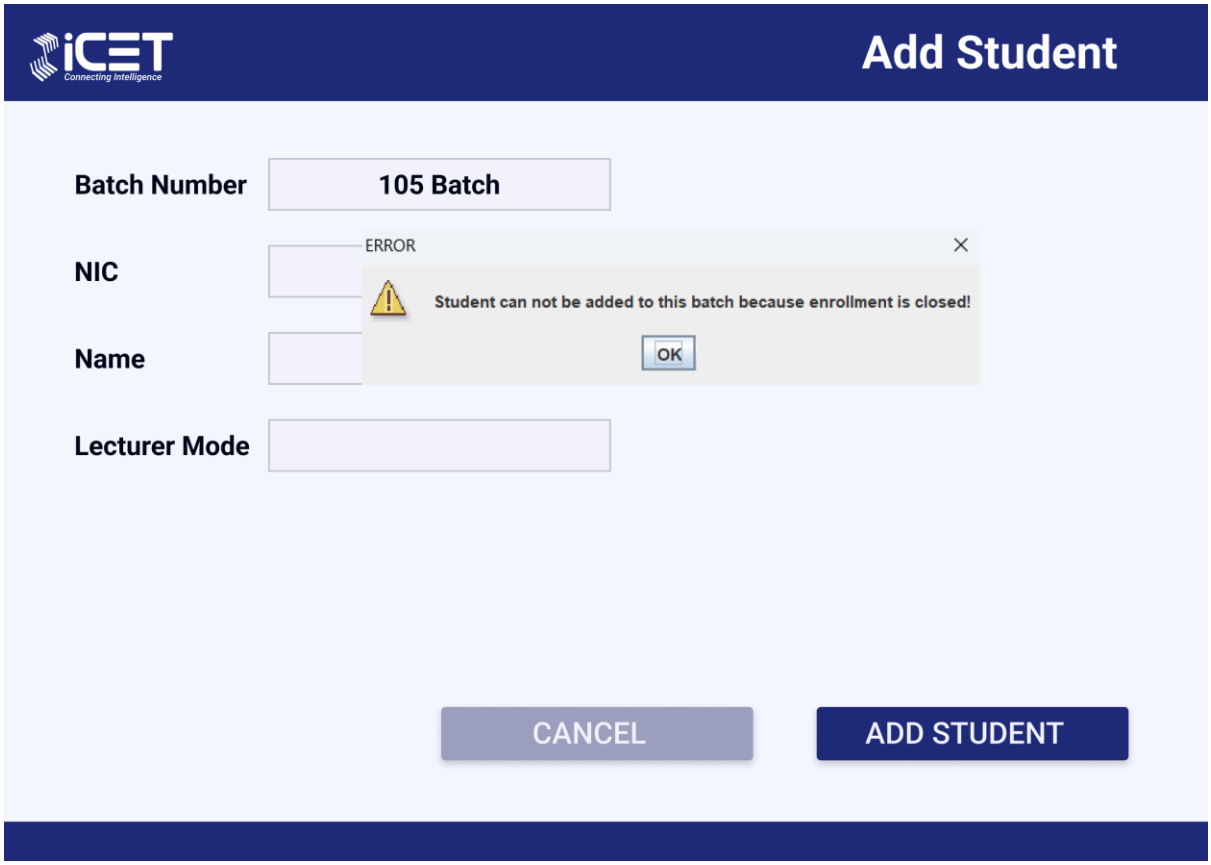


Figure 5 - Warning(Enrollment Closed)

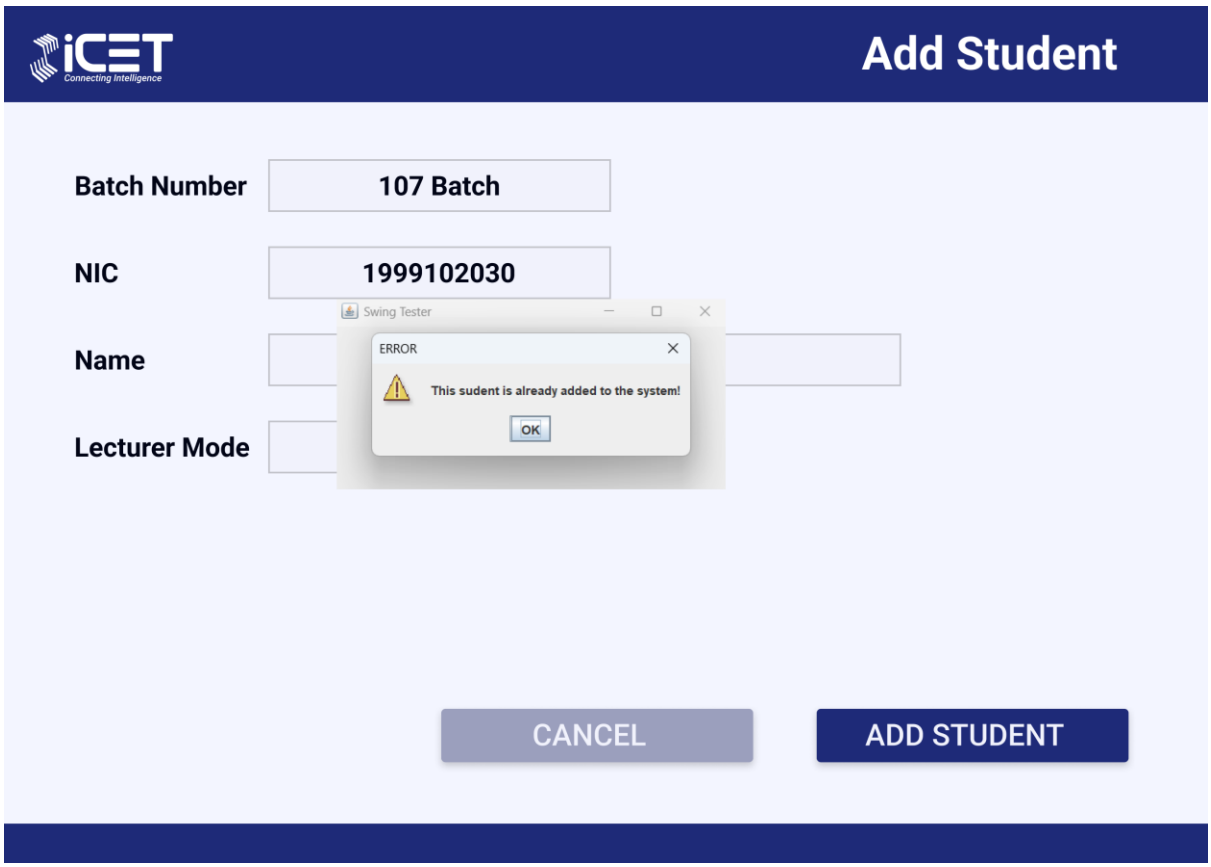


Figure 6 - Warning(Already added student)

After the student details are entered successfully, the user should click the "Add Student" button to add the student to the system. Once added, the system should display the student registration number along with a "Student added successfully" message. The user can then click the "Back to Home Page" button to return to the home page.

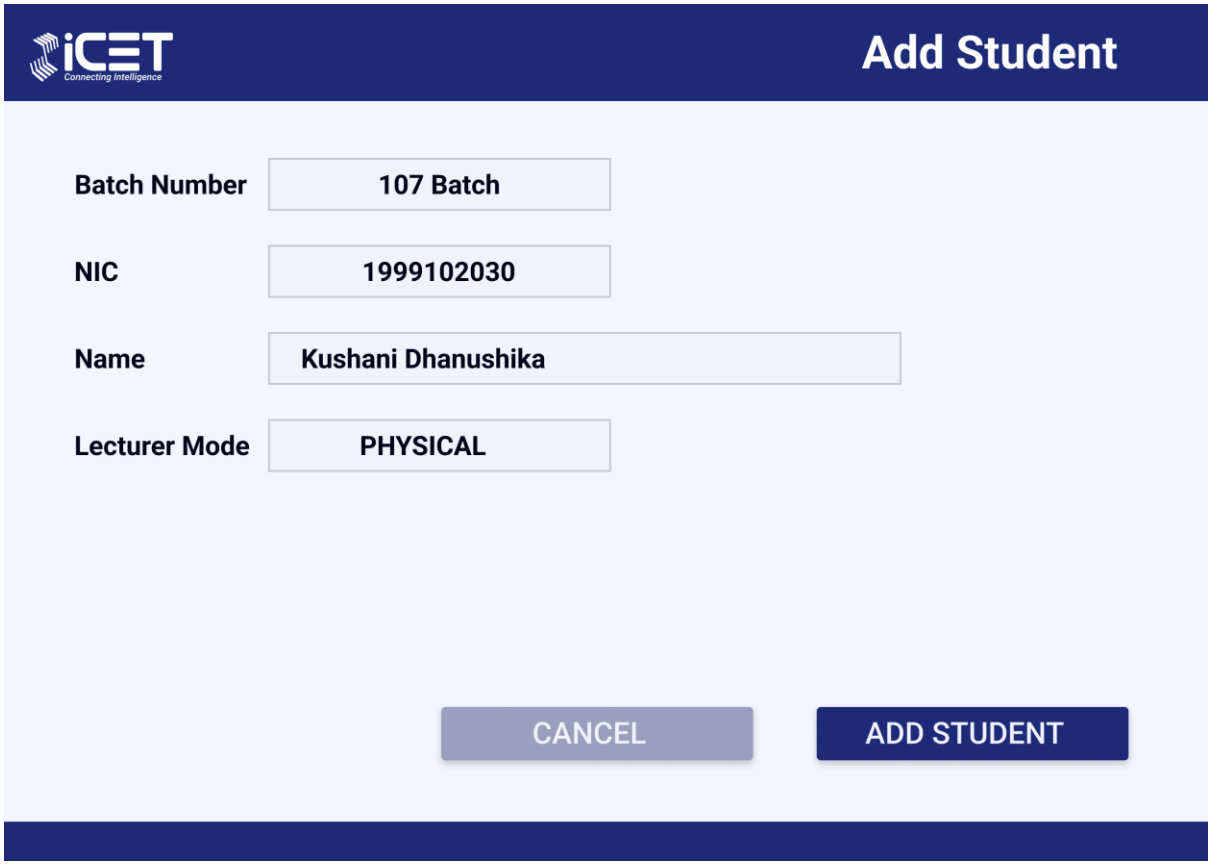


Figure 7 - Add New Student Details

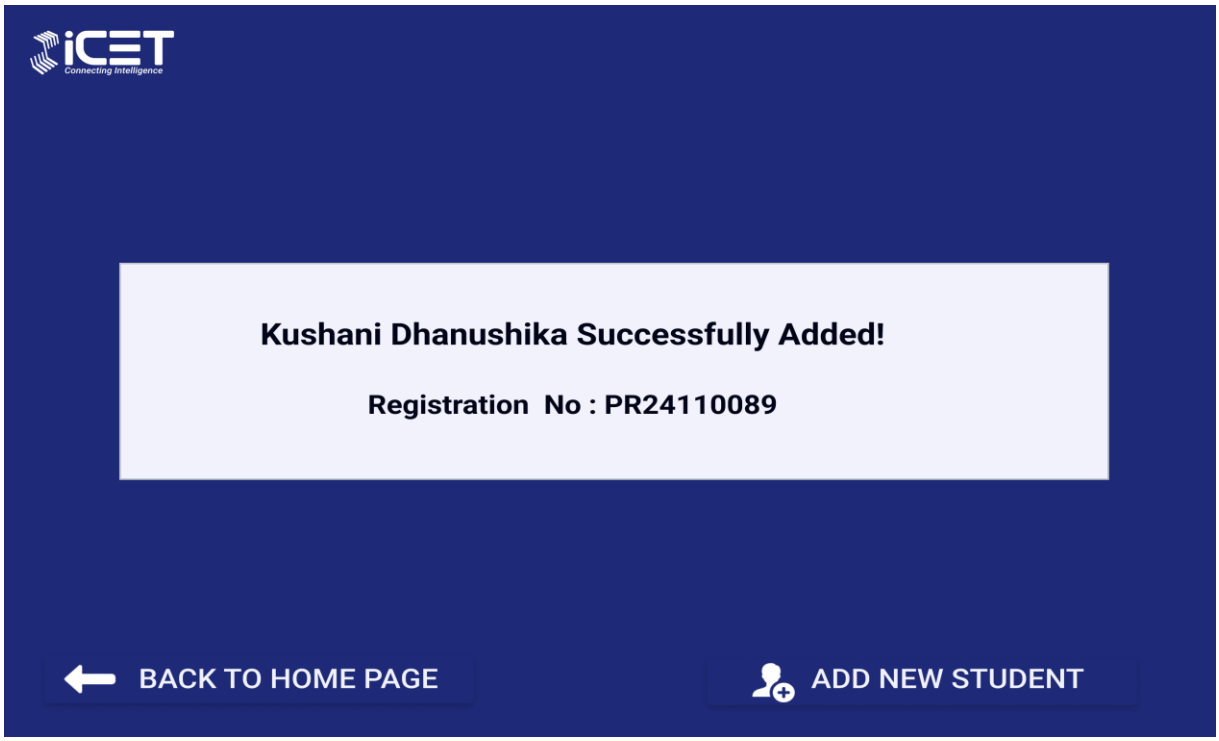


Figure 8 - Add New Student Successfully

1.2 Update Student

With this option, the user can update previously added student details (only the Name and NIC). First, the user needs to find the student by entering the Student ID. After the Student ID is entered, the system should check whether it exists in the system. If the Student ID has not been added yet, a message should be displayed as “This student not exist in the system”.

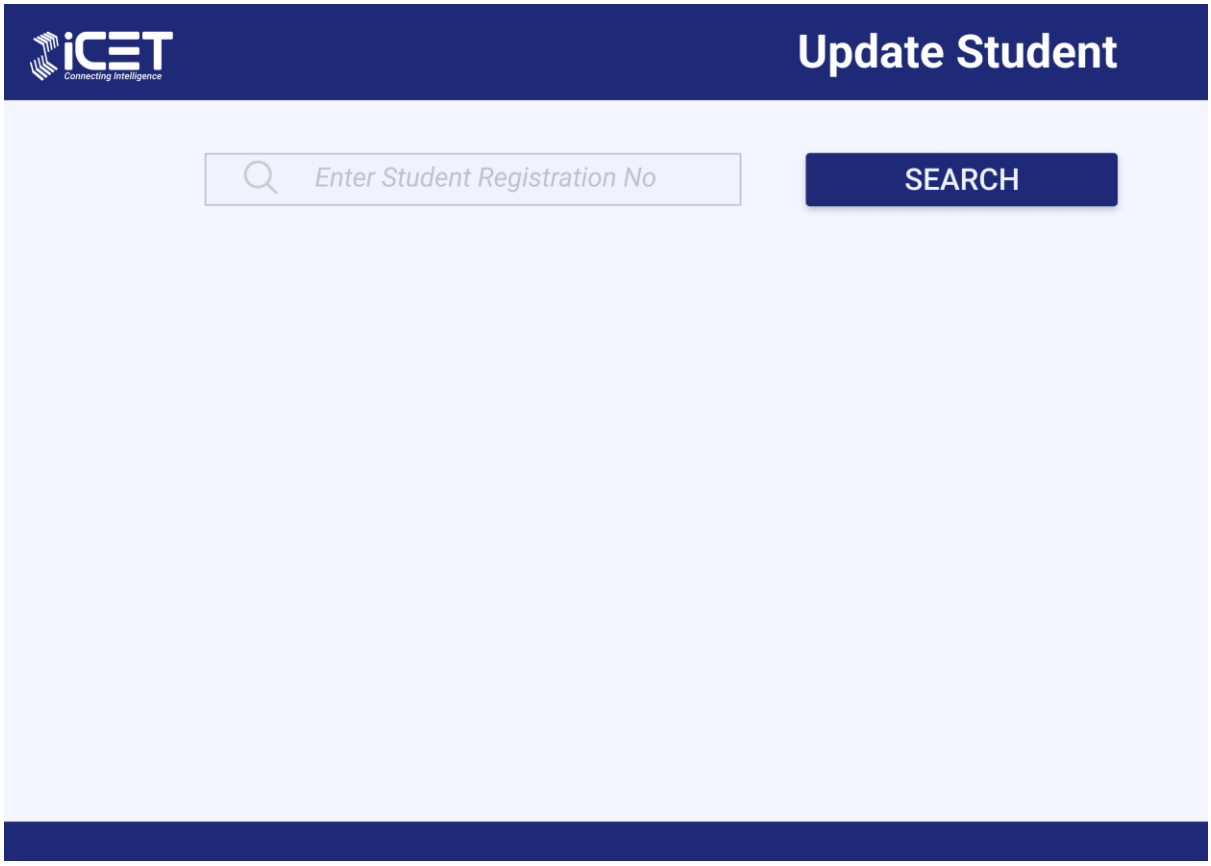


Figure 9 - Update Student

Once the user has entered a valid Student ID, the system will display the current details, including the Student ID, Student Name, and Student NIC. After displaying the student details, the user can update either the Student Name or the Student NIC.

If the user change Student NIC, The system should check whether the entered NIC already exists in the system. If the NIC is found, the system should display a message indicating that the NIC already exists. If the NIC is not found, the system should allow the user to update the Student NIC.

Once the update has been completed successfully, the system should display a success message.

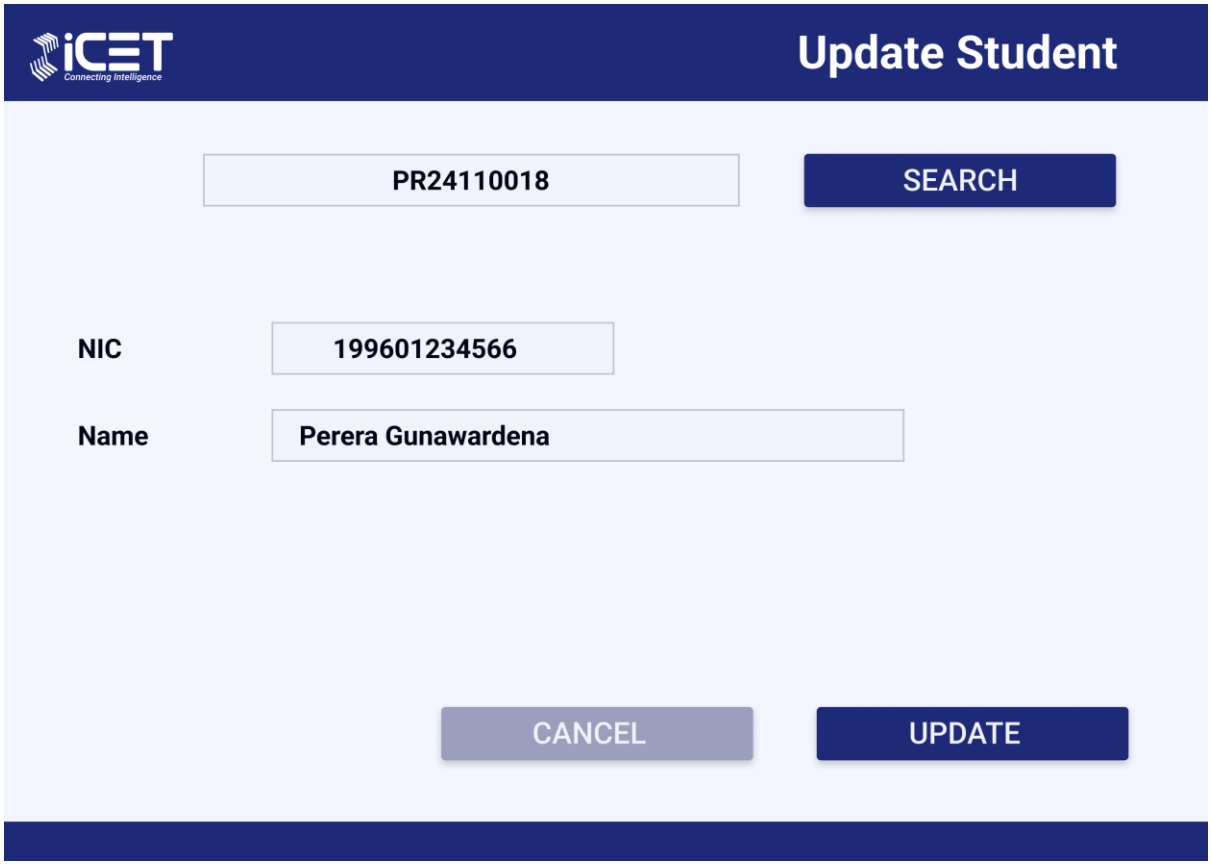


Figure 10 - Update Name or NIC

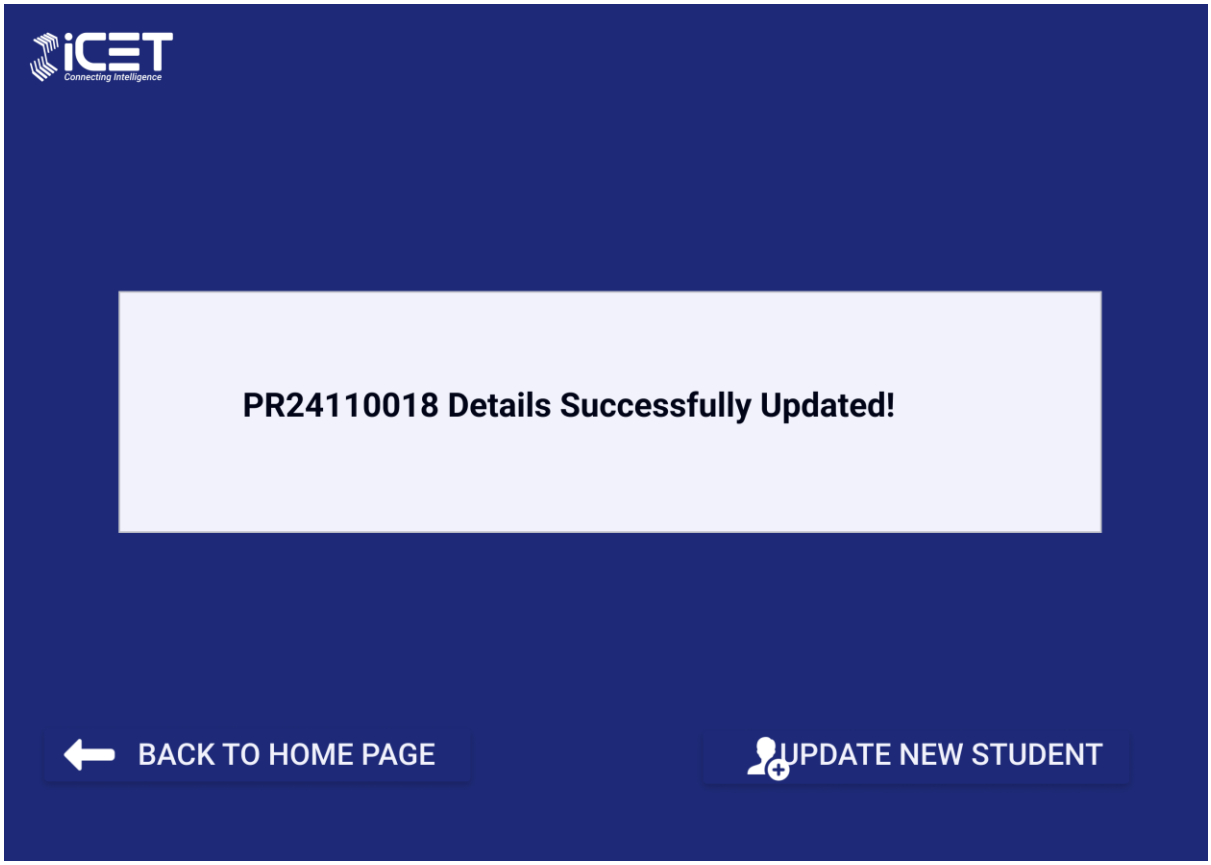


Figure 11 - Update Successfully

1.3 View Student Profile

With this option, the user can view student details. First, the user must enter a valid Student ID. After the Student ID is entered, the system should check whether it exists in the database. If the Student ID has not been added yet, a message should be displayed as "This student does not exist in the system". If the Student ID is valid, the system should display the student details as shown in Figure 13,14 and 15.

The user can view 6 details about the student in this option: Registration No, Name, NIC, PRF Marks, DBMS Marks, and Student GPA. The Student GPA should be calculated based on the marks, using the GPA values mentioned in Figure 12. The system should calculate the GPA as:

(PRF Marks GPA + DBMS Marks GPA) / 2

Some students may be absent (indicated by -1) from the exam, and for some students, the exam may not have been conducted yet (indicated by -2).

Marks Range	GPA Value
90 – 100	4.25
80 – 89	4.00
75 – 79	3.70
70 – 74	3.30
65 – 69	3.00
60 – 64	2.70
55 – 59	2.30
50 – 54	2.00
45 – 49	1.70
40 – 44	1.30
30 – 39	1.00
20 – 29	0.70

Figure 6- GPA Value

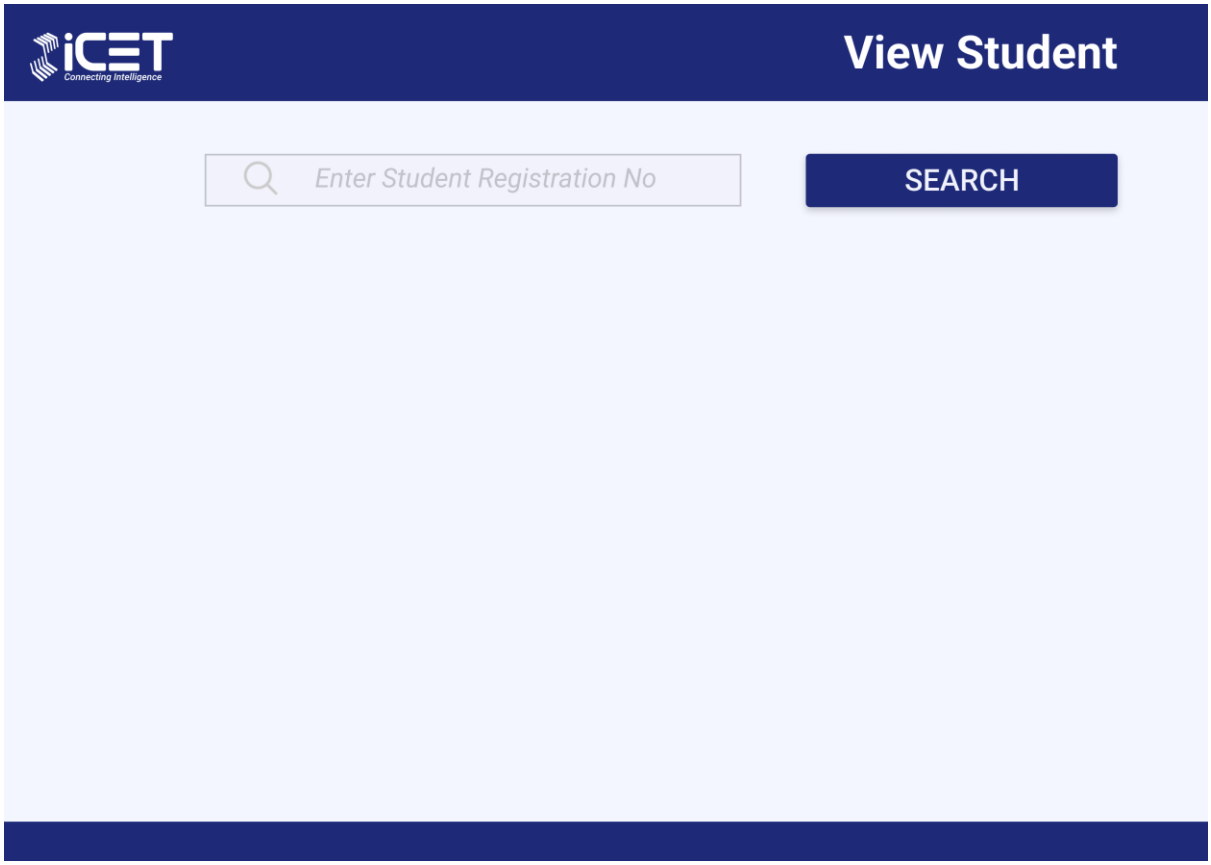


Figure 127 - View Student Profile

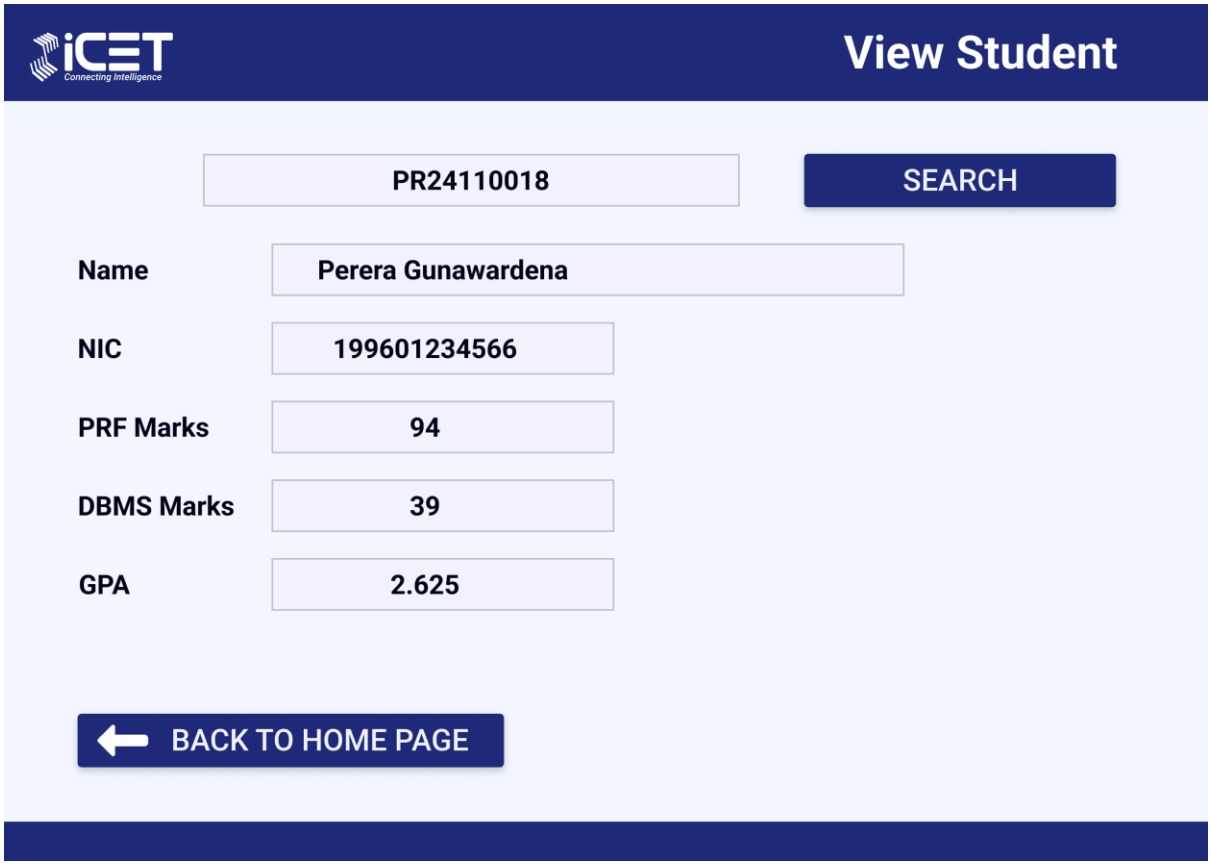


Figure 13 - Student Profile Data

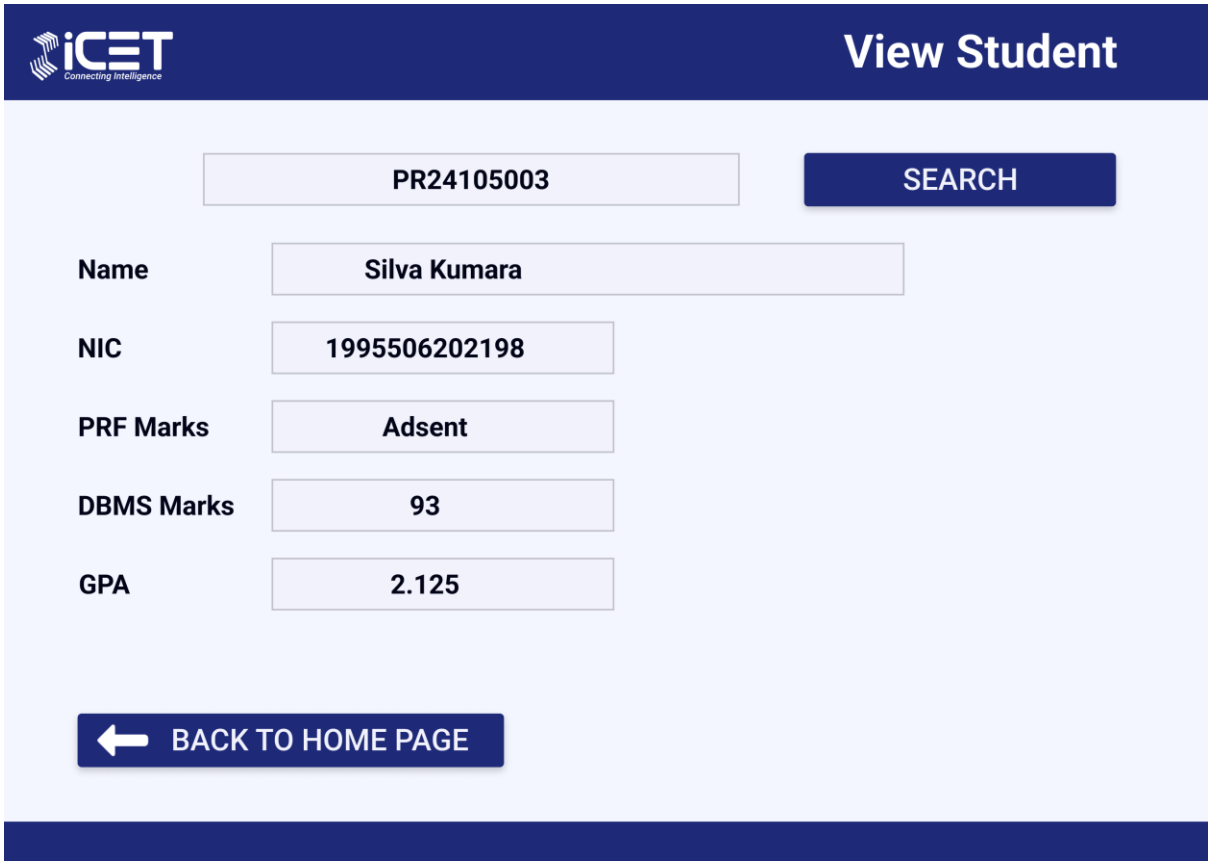


Figure 14 - Student Profile Data (Absent)

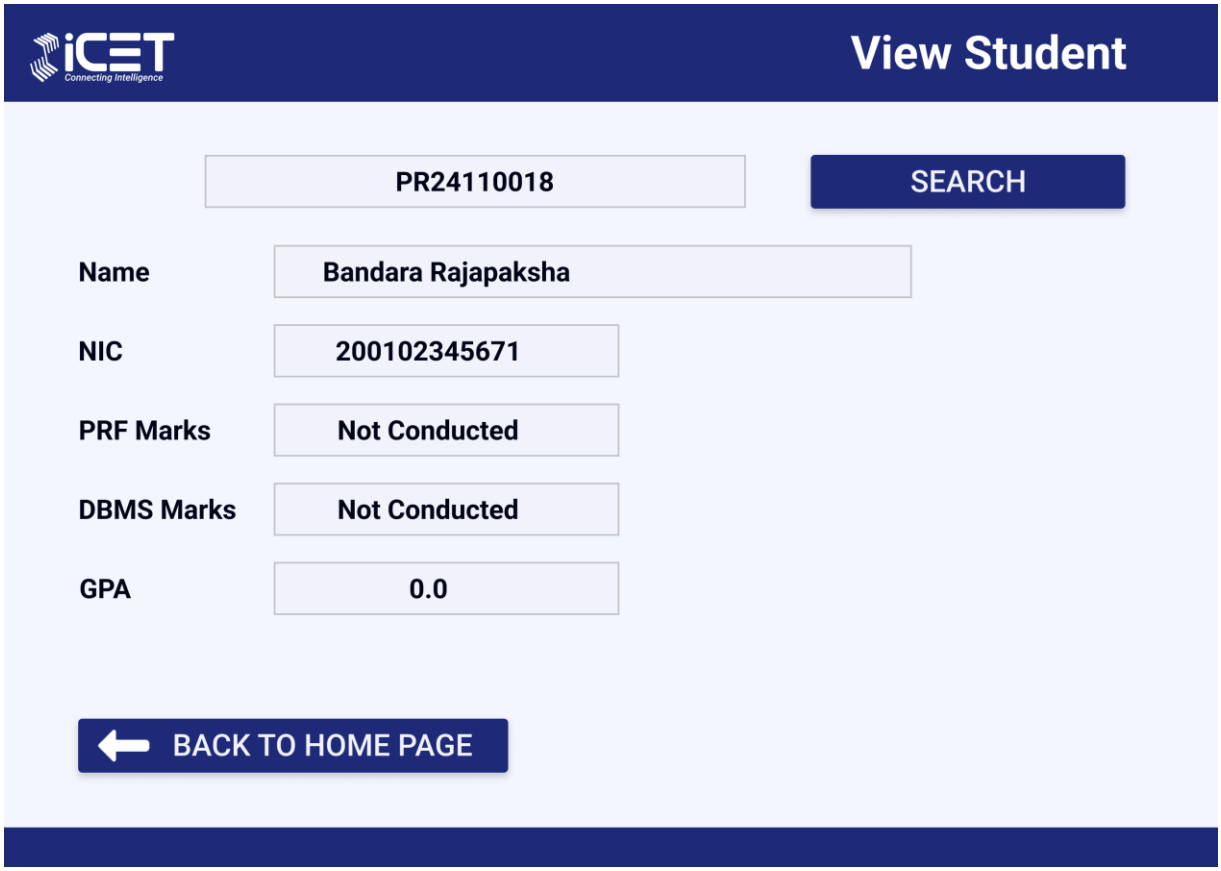


Figure 15 - Student Profile Data(Not Conducted Exam)

1.4 Delete Student Profile

With this option, a student can be deleted from the system. First, the user needs to enter the valid Student ID; otherwise, it should handle like previously. Once the user has entered a valid student id, the student can be deleted.

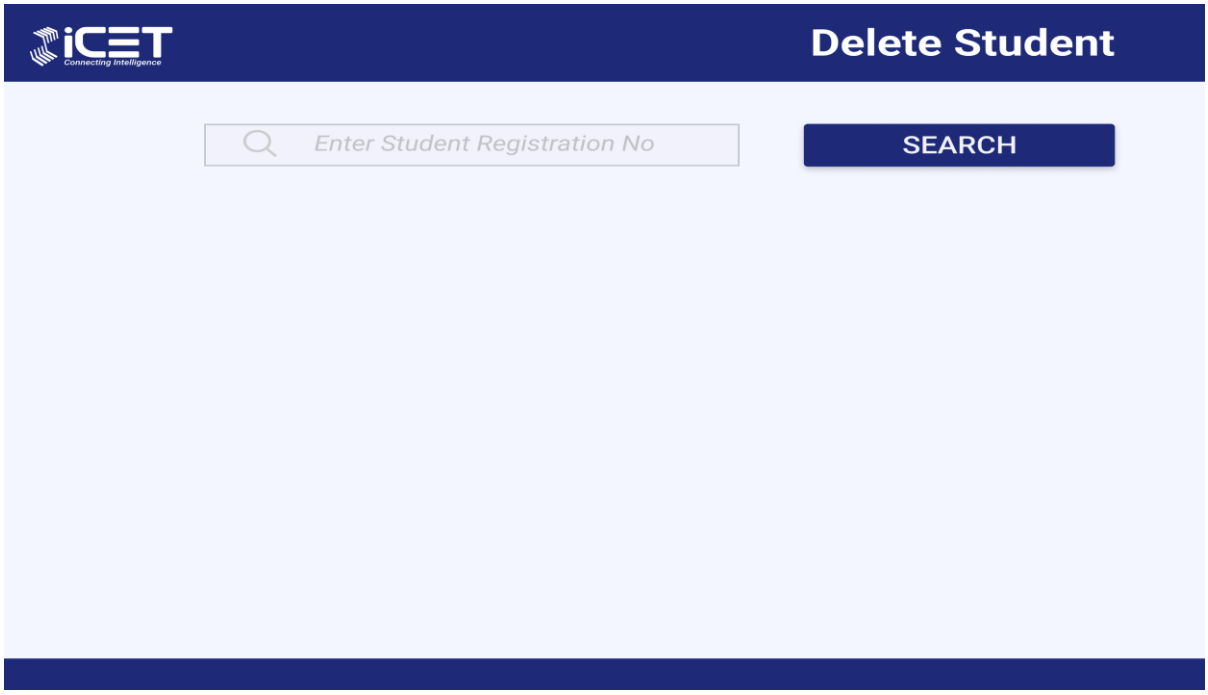


Figure 168 - Delete Student

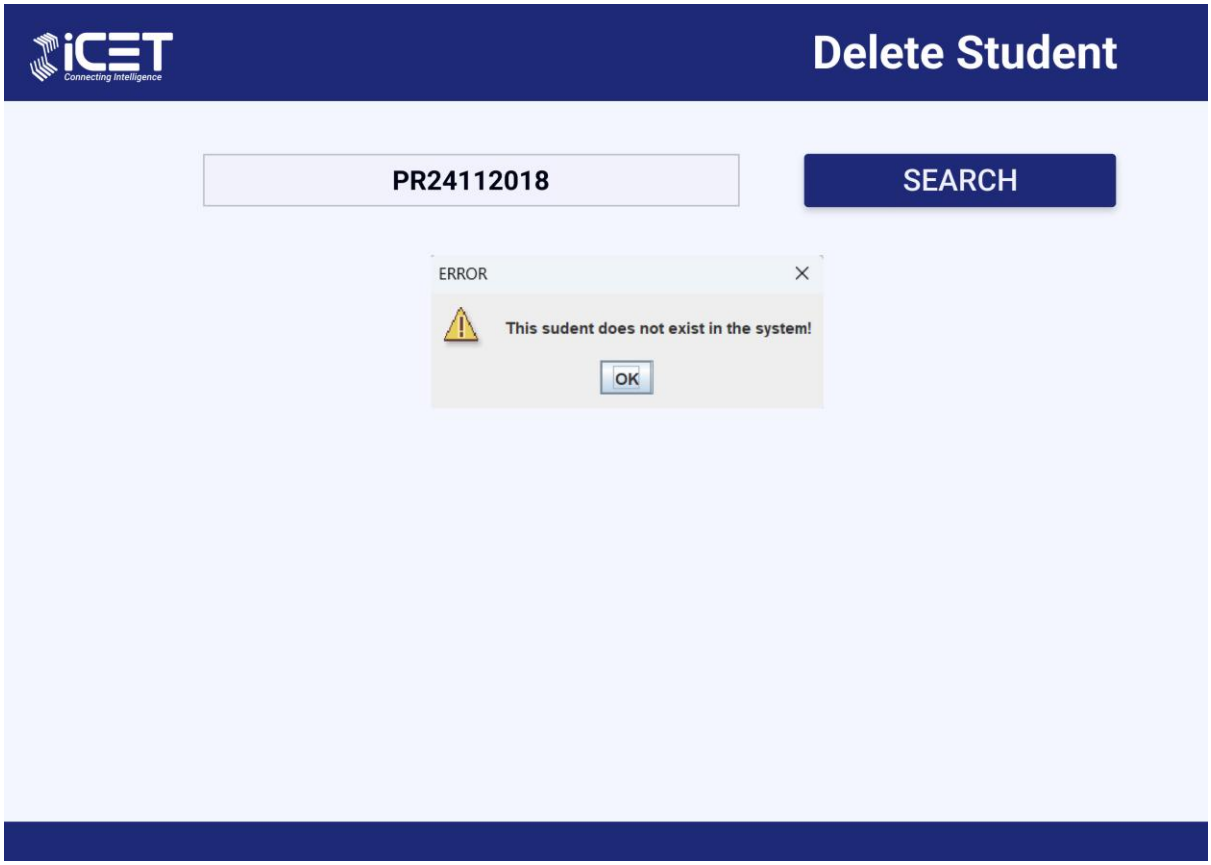


Figure 96 - Warning(Not Existing in the system)

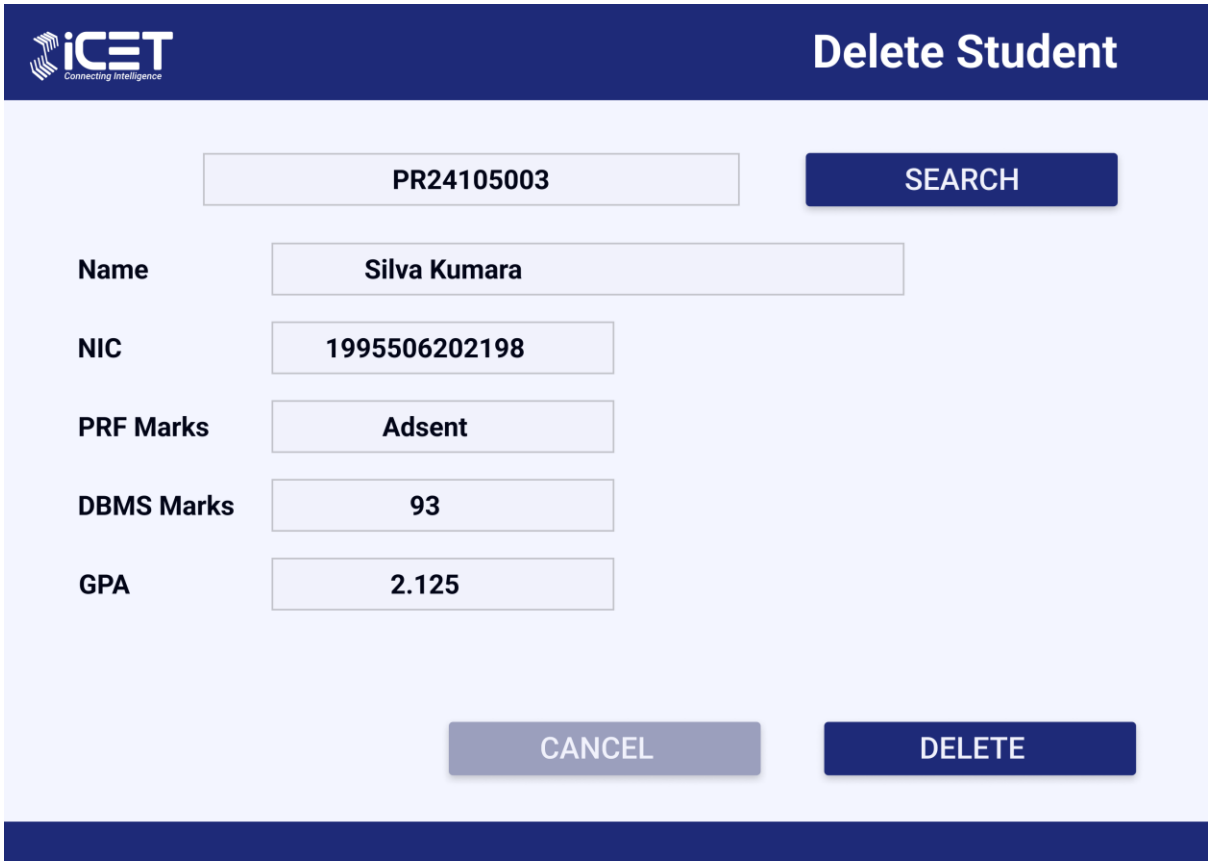


Figure 1710- Display Student data before delete

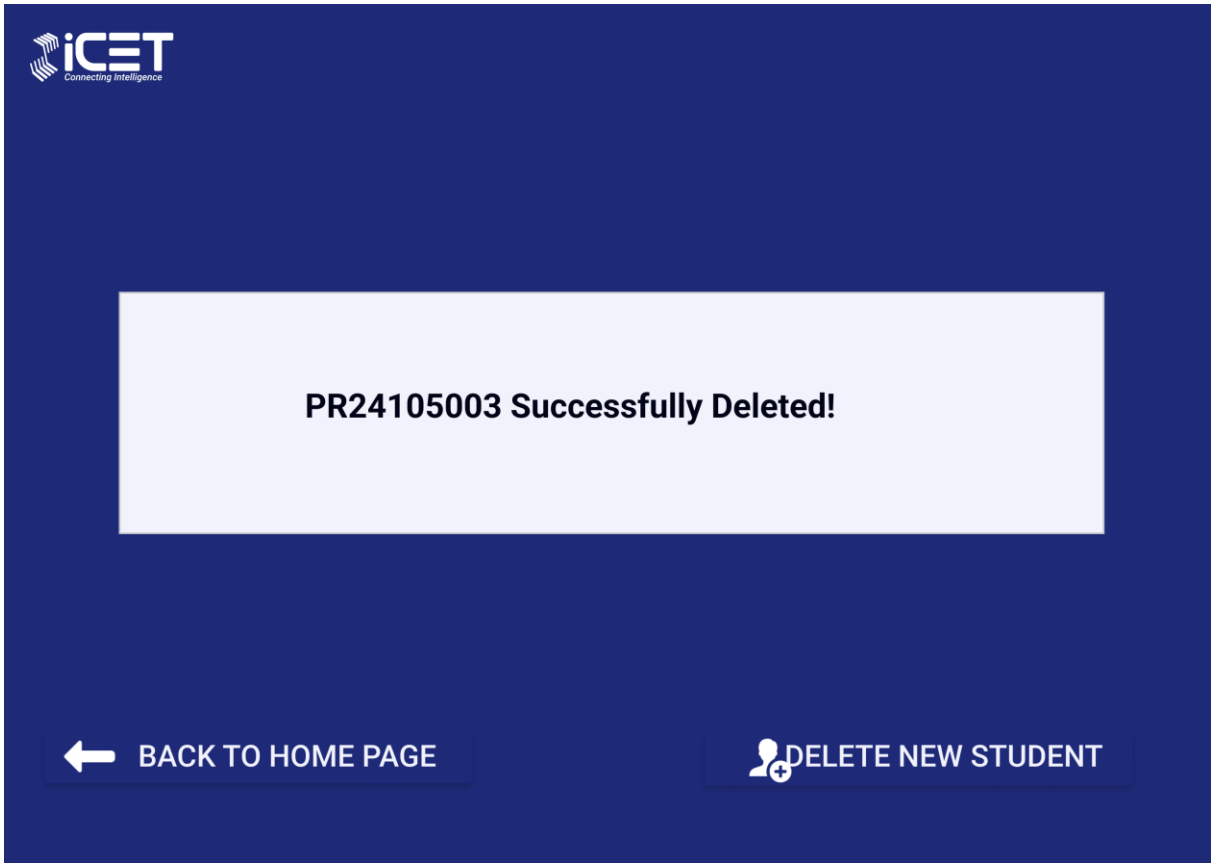


Figure 18 - Delete Successfully

[2] Batch Management

There are three options in Batch Management, and the user can select the corresponding option they wish to execute. Below is the Batch Management interface you should develop.



Figure 1911 - Batch Management

2.1 Add Batch

The system stores three key details for each batch:

- Batch ID – Each batch has a unique number starting from 100.

- Batch Status – There are two possible statuses for a batch. The initial status of the batch is ENROLLMENTOPEN:
    - ENROLLMENTOPEN = 1
    - ENROLLMENTCLOSED = 0

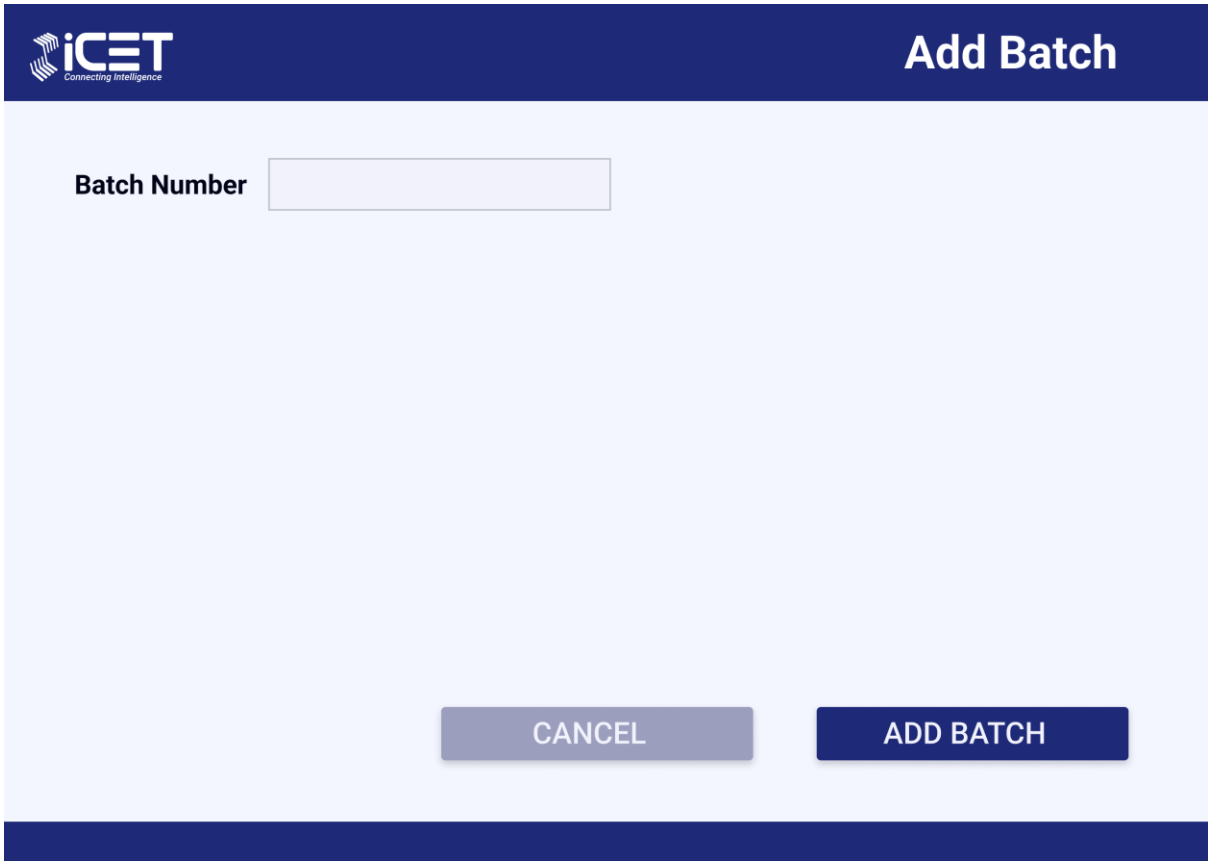


Figure 20 - Add Batch

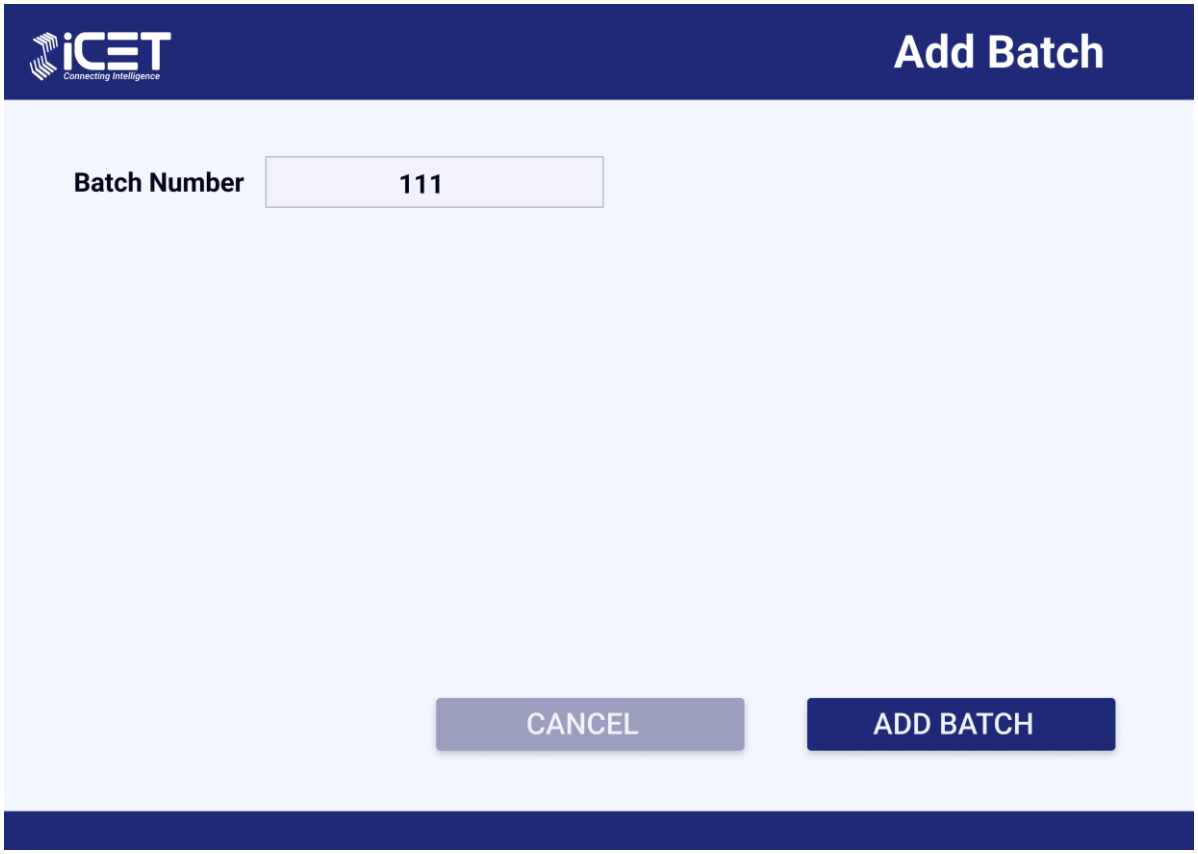


Figure 21 - Add Batch Number

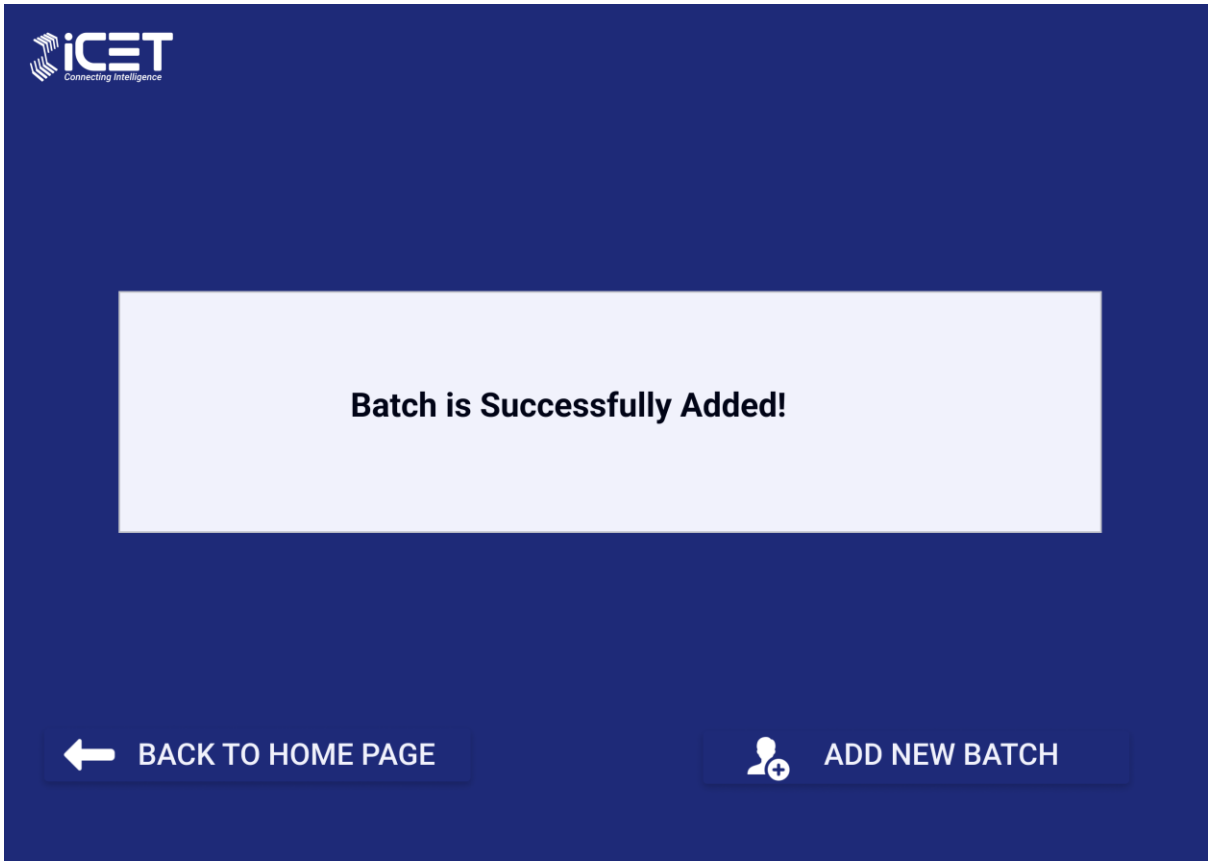


Figure 22 - Successfully add batch

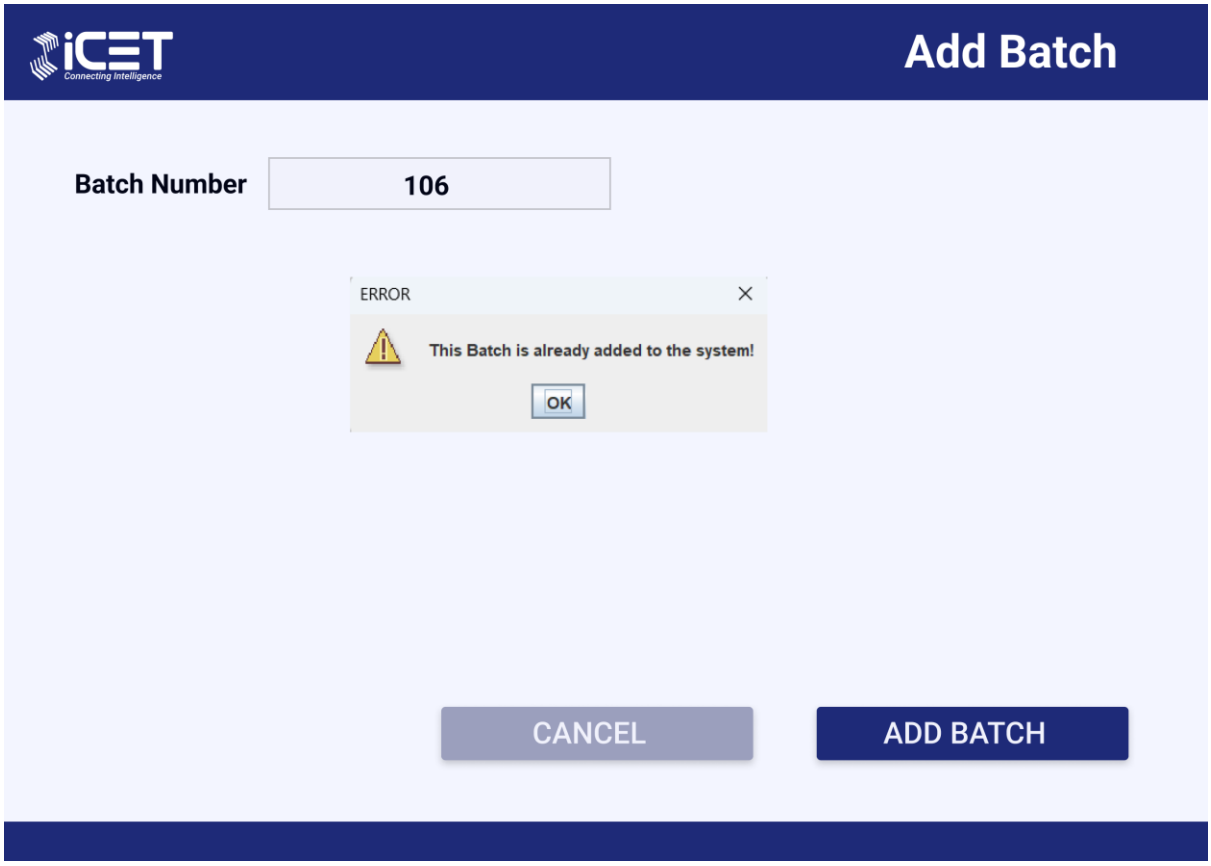


Figure 23 - Warning(Already added in the system)

2.2 Update Batch

With this option, the user can update previously added batch details (only the batch status). First, the user needs to find the batch by entering the batch ID. After the batch ID is entered, the system should check whether it exists in the system. If the batch ID has not been added yet, a message should be displayed: “This batch does not exist in the system.

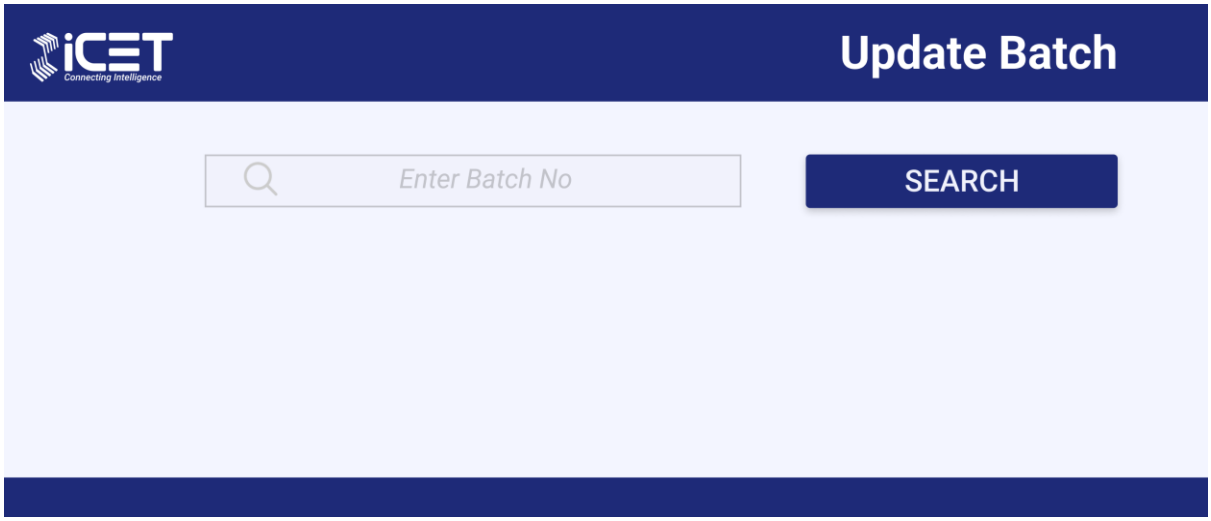


Figure 24 - Update Batch

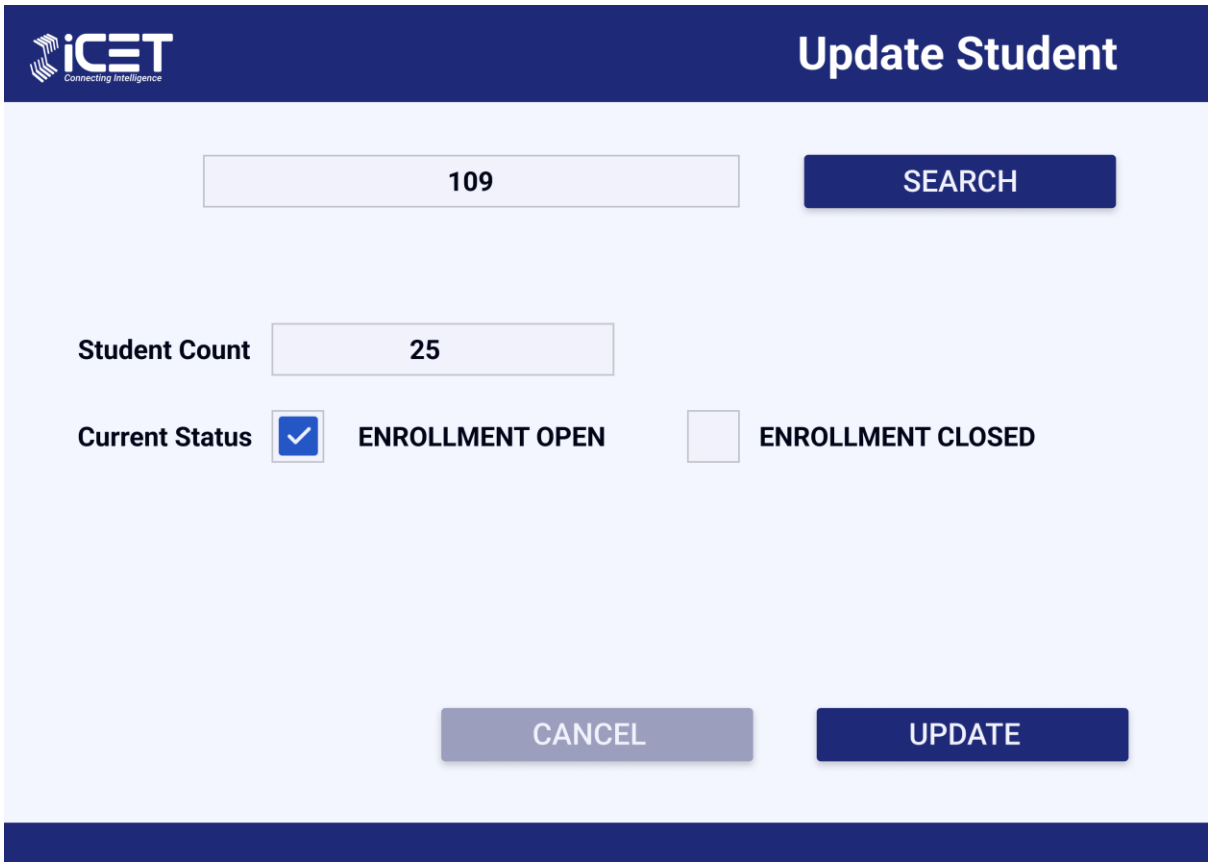


Figure 25 – Show current details of the batch

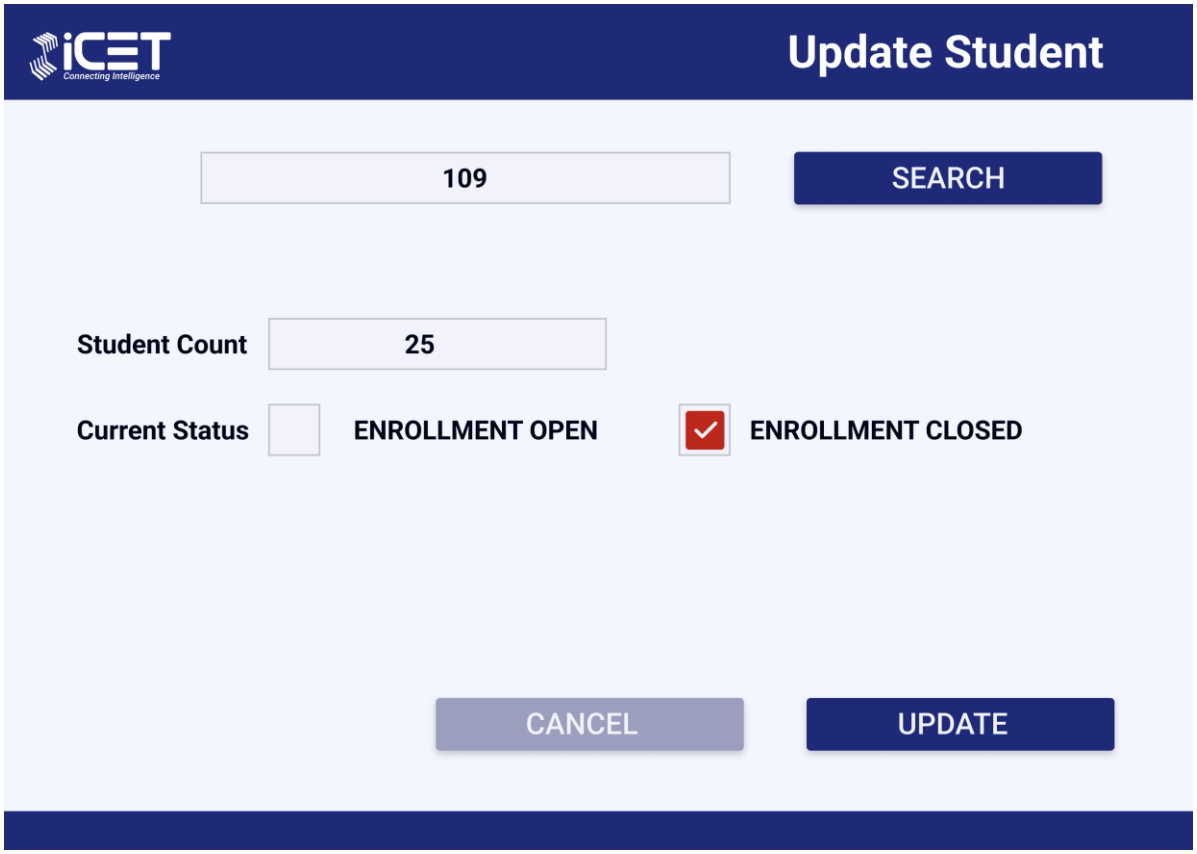


Figure 26 – Change the ststus of the batch

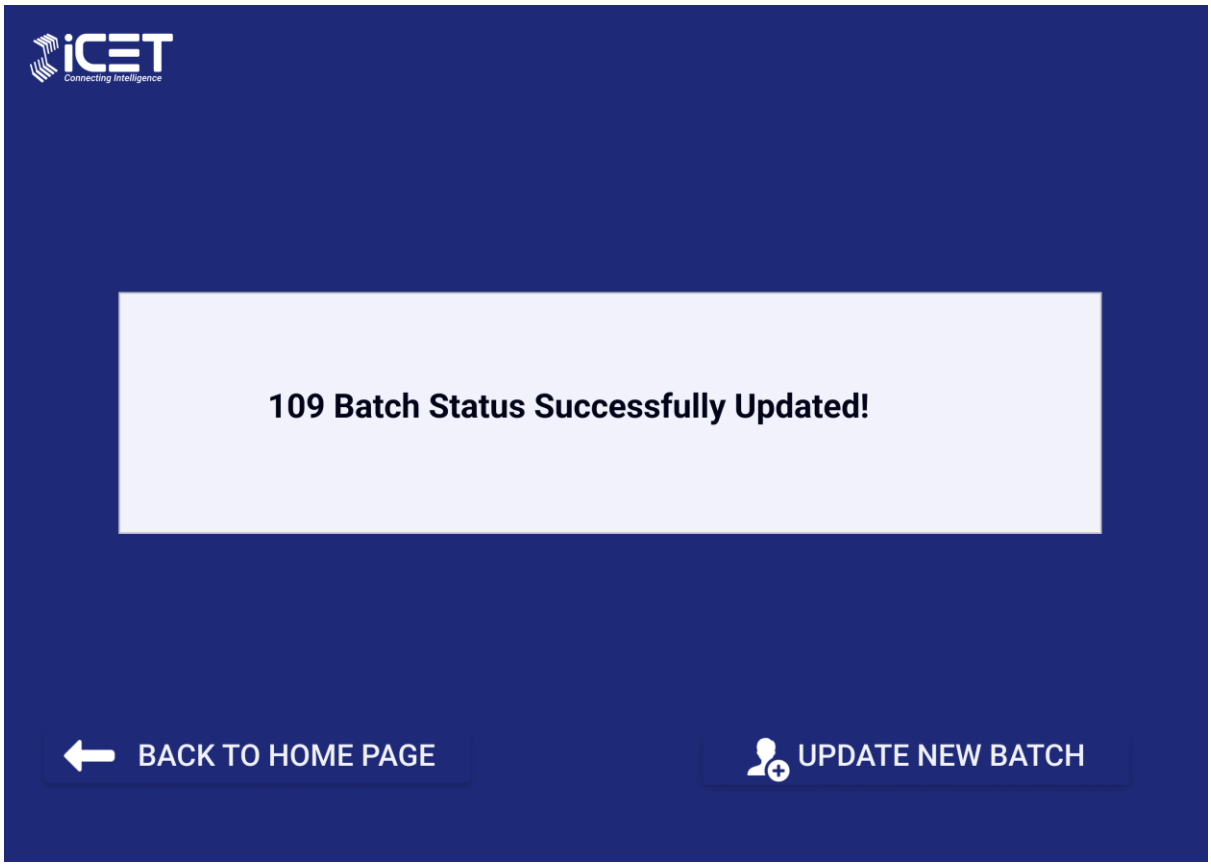


Figure 27 - Successfully change batch status

2.3 View Batches

This displays all the batches with details, including the batch name, student count, and status.

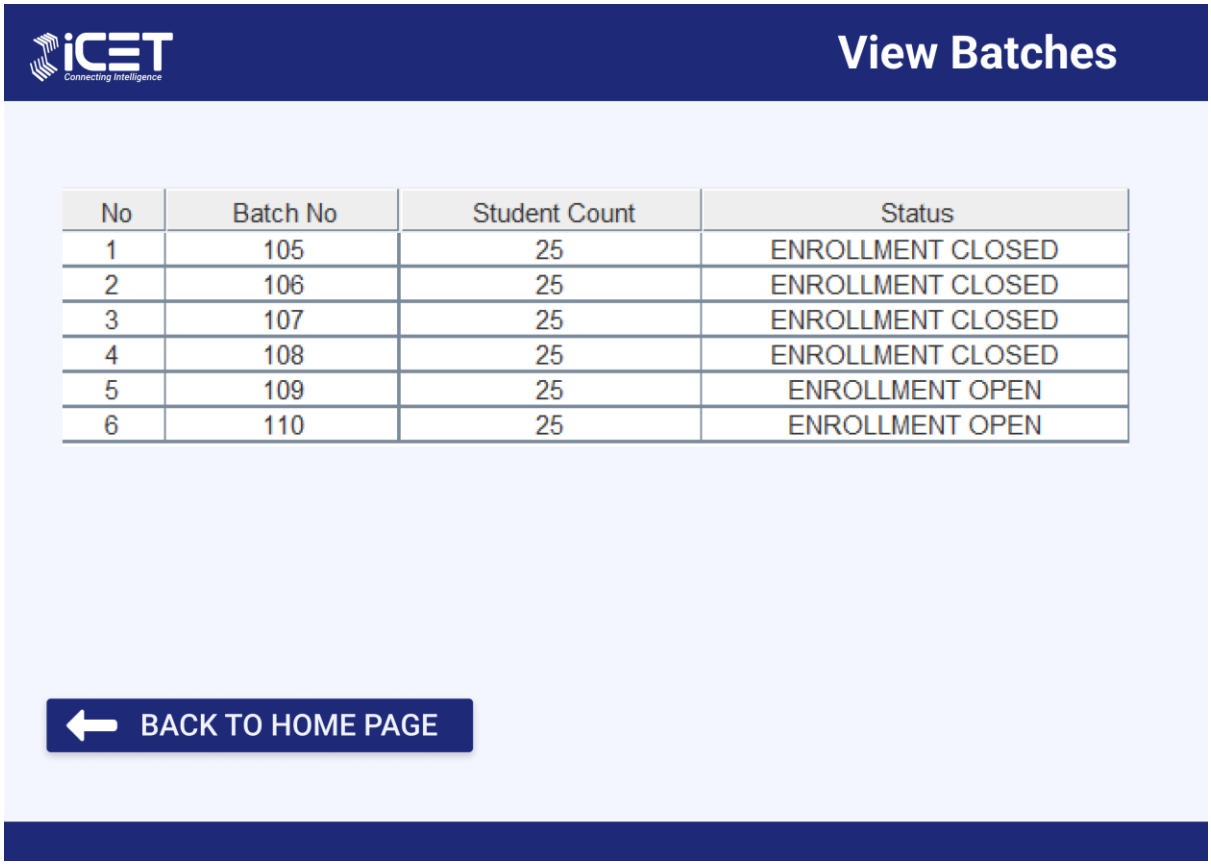


Figure 28 - View Batch Data

[3] Grade Management

There are two options in Grade Management, and the user can select the corresponding option they wish to execute. Below is the Grade Management interface you should develop.



Figure 2912 - Grade Management

3.1 PRF Marks Update

First, the user needs to find the student whose PRF marks should be updated via Student ID. (Invalid Student ID's should be handled like previously) Once the user has entered a valid student ID, it is going to display the current details. Then user can add new marks. Again, marks should be validated here.

The marks format is as follows: if the student is absent, the mark should be **-1**; otherwise, marks must be within the range of **0 to 100**.

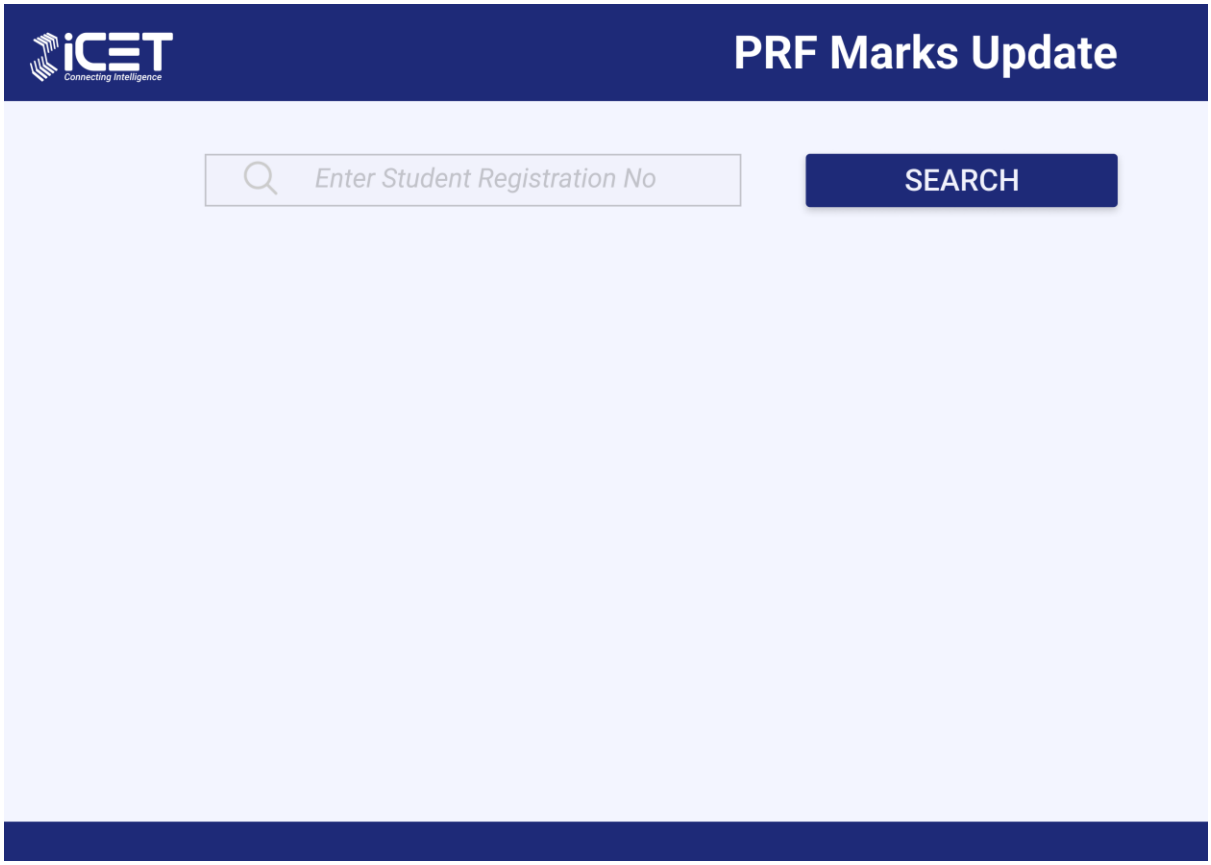


Figure 3013 - PRF Marks Update

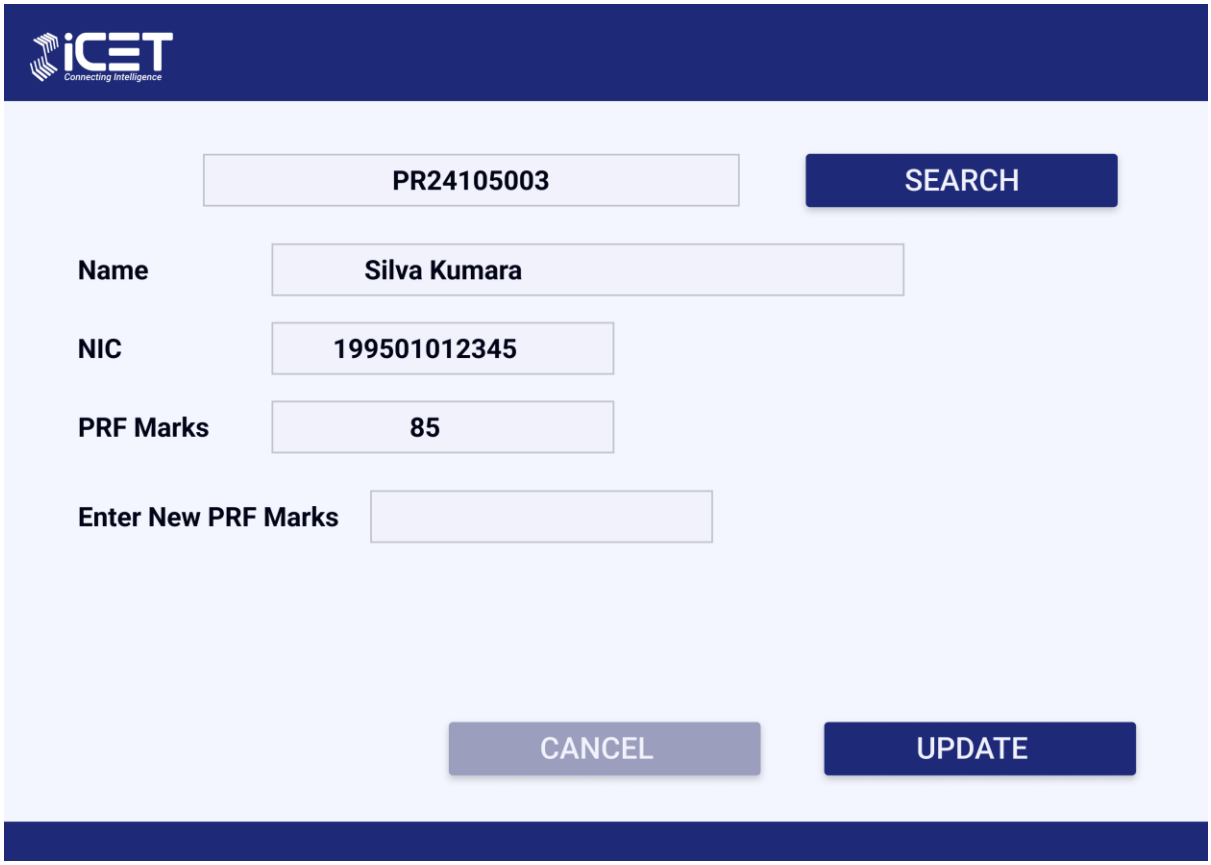


Figure 31 - Show current PRF Marks

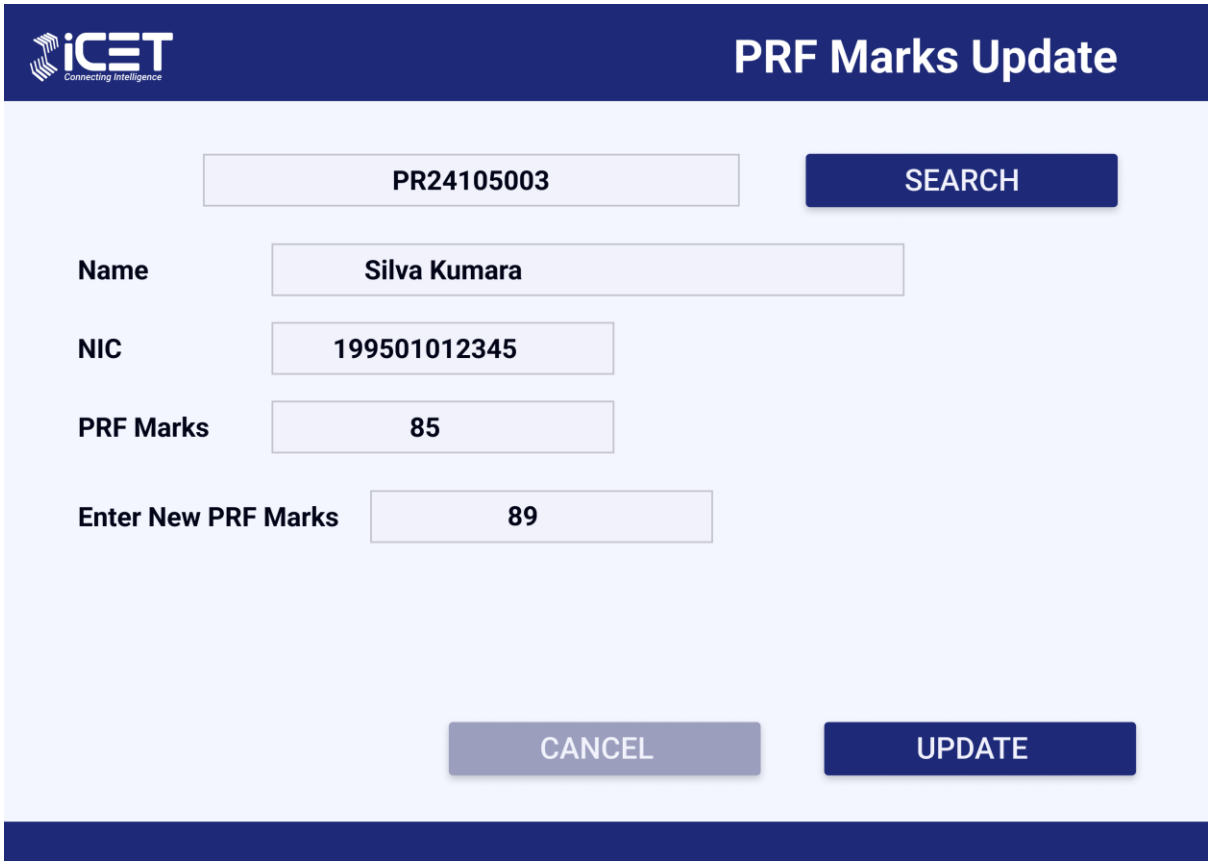


Figure 32 – Enter New PRF Marks

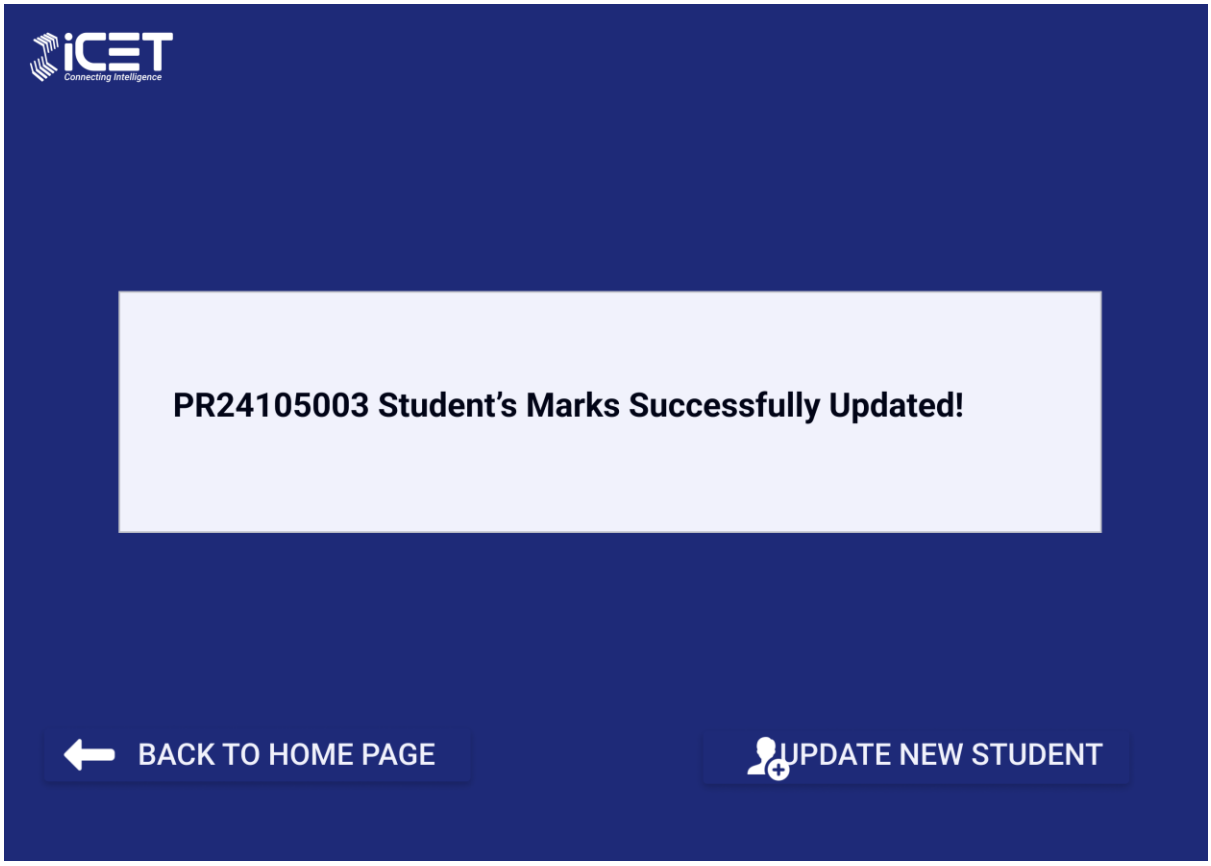


Figure 33 - Update new marks successfully

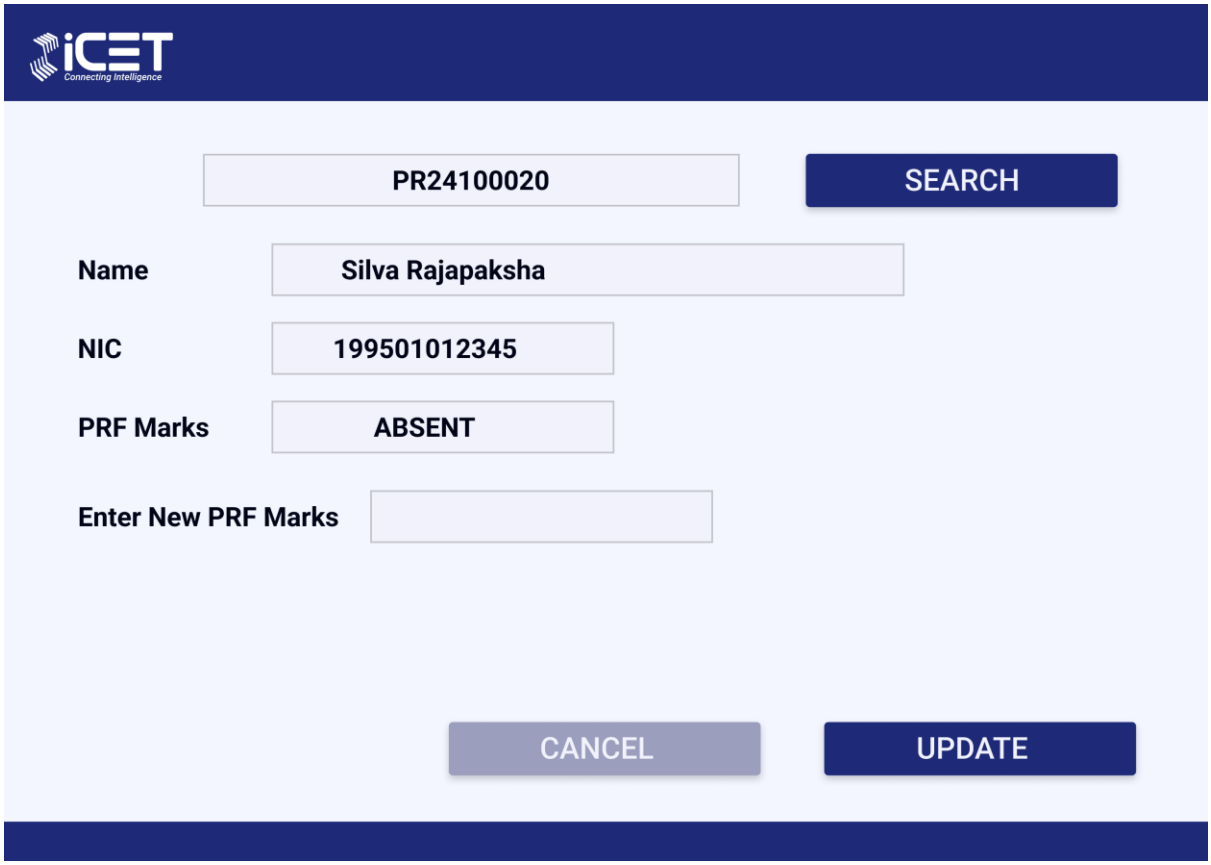


Figure 34 - Absent mark update

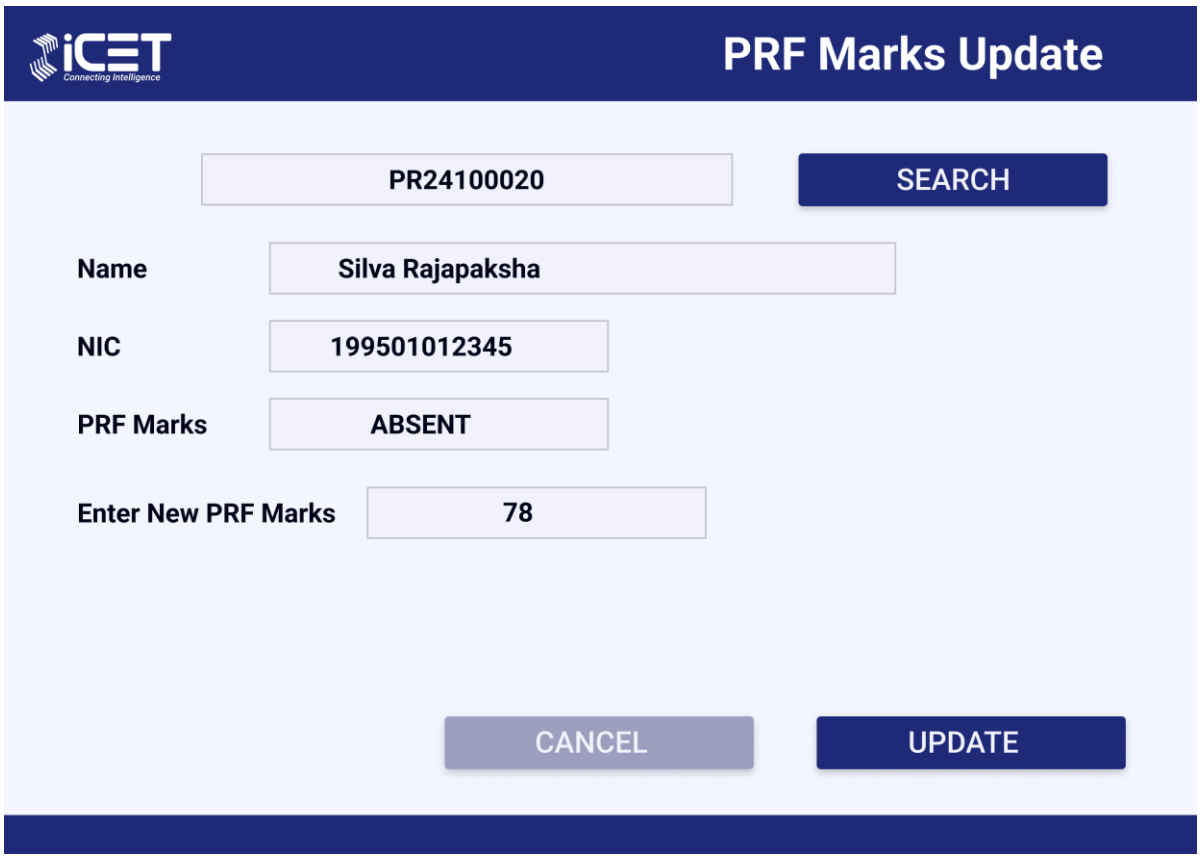


Figure 35 – Add New Marks

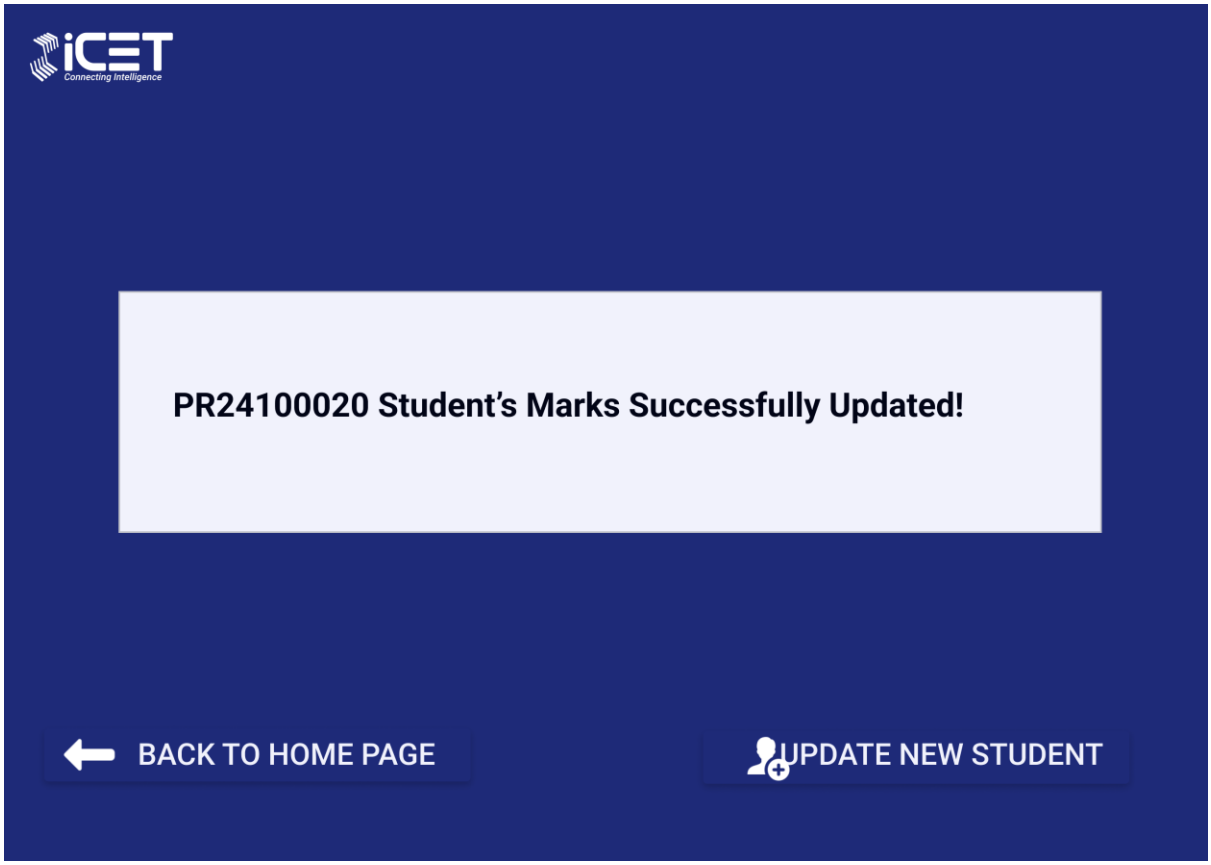


Figure 36 - Absent mark update successfully

3.2 DBMS Marks Update

First, the user needs to find the student whose DBMS marks should be updated via Student ID. (Invalid Student ID's should be handled like previously) Once the user has entered a valid student ID, it is going to display the current details. Then user can add new marks. Again, marks should be validated here.

The marks format is as follows: if the student is absent, the mark should be **-1**; otherwise, marks must be within the range of **0 to 100**.

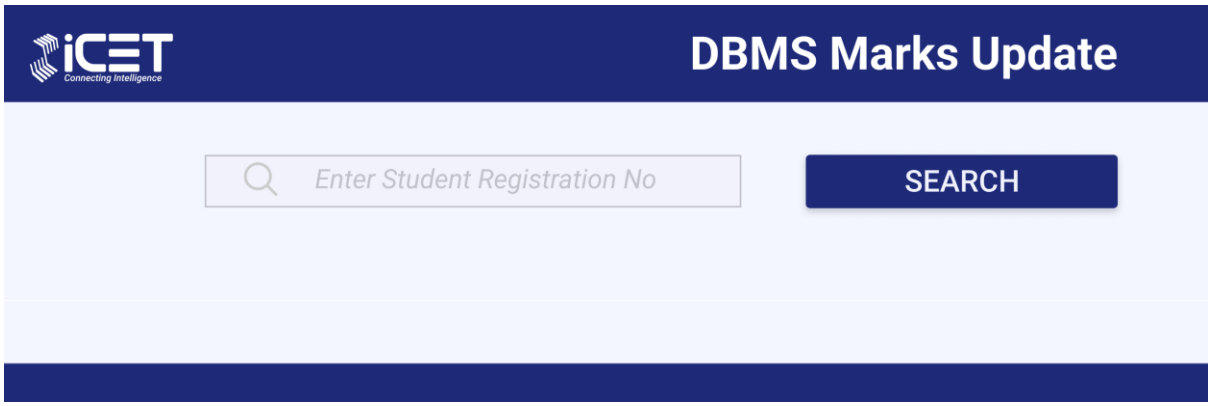


Figure 37 - - DBMS Marks Update

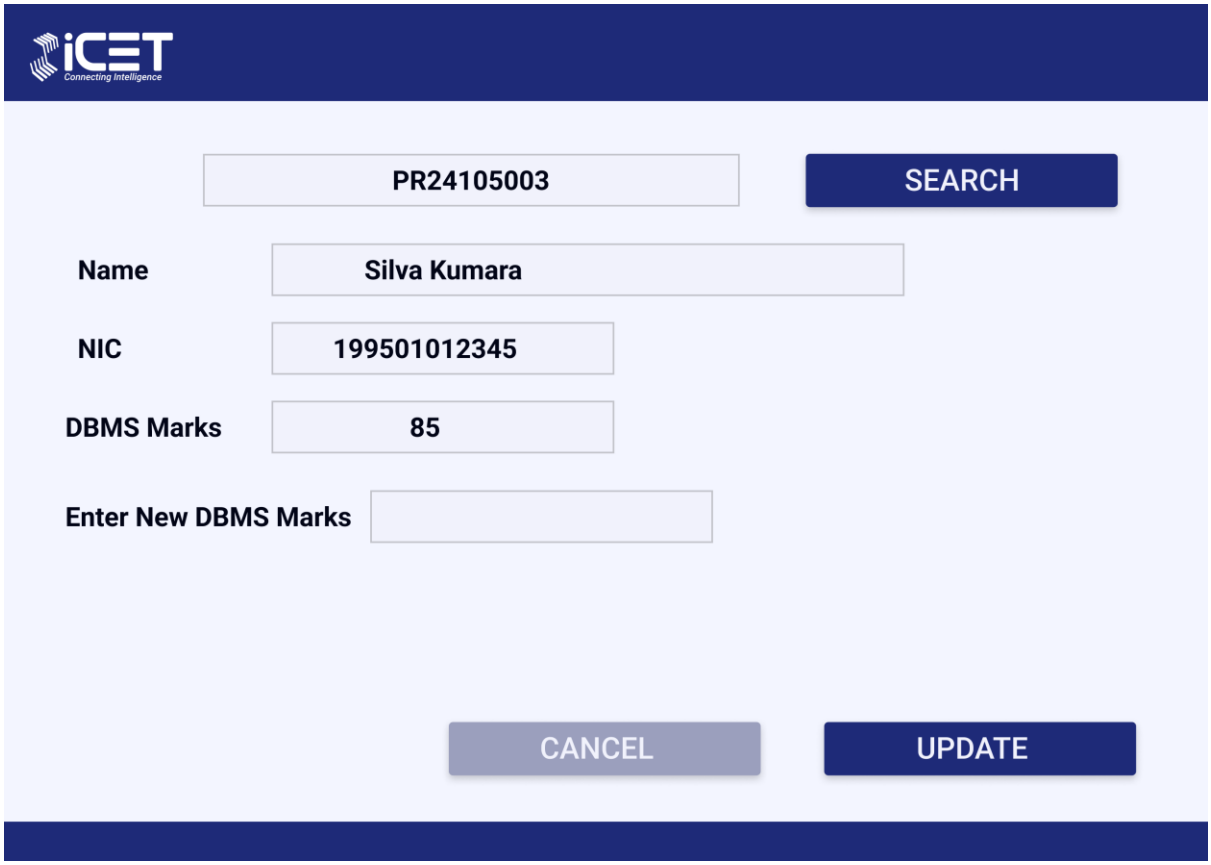


Figure 38 - Show current DBMS Marks

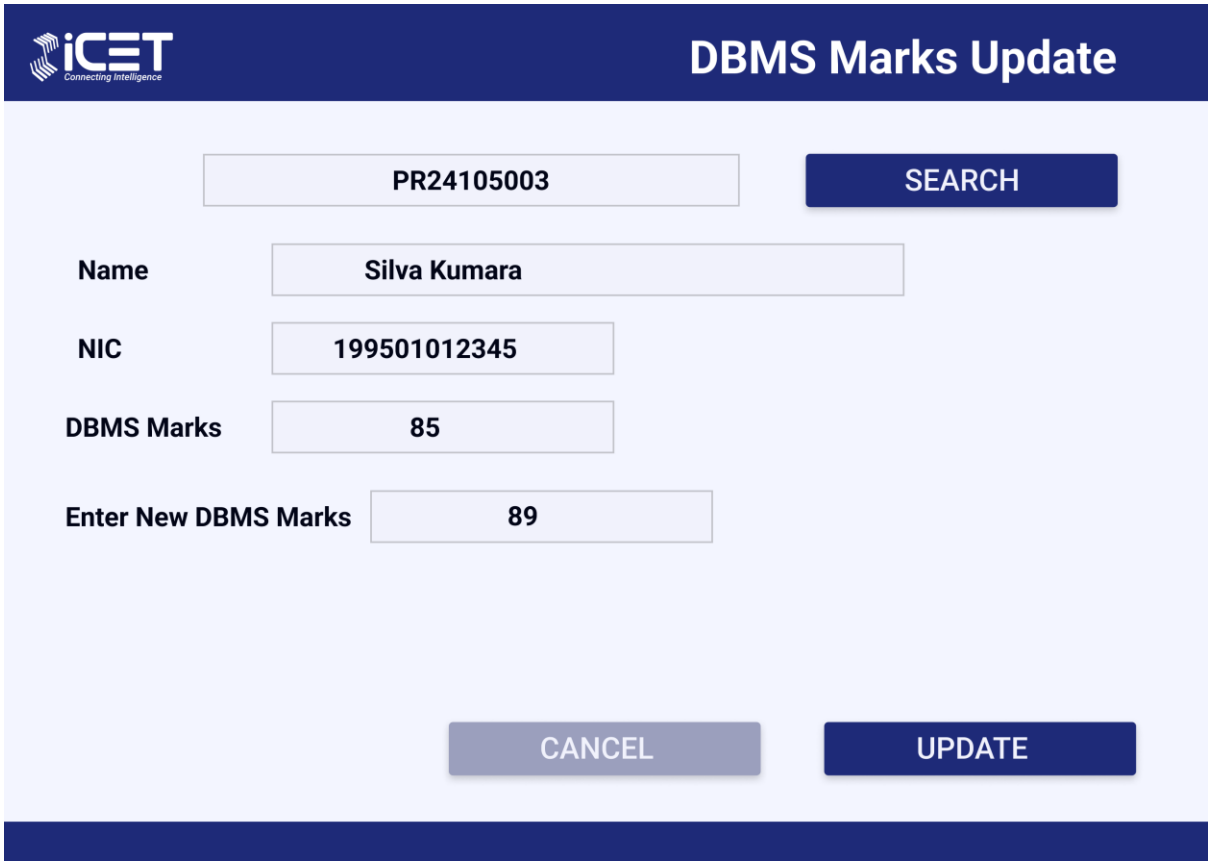


Figure 39 - Update already added mark

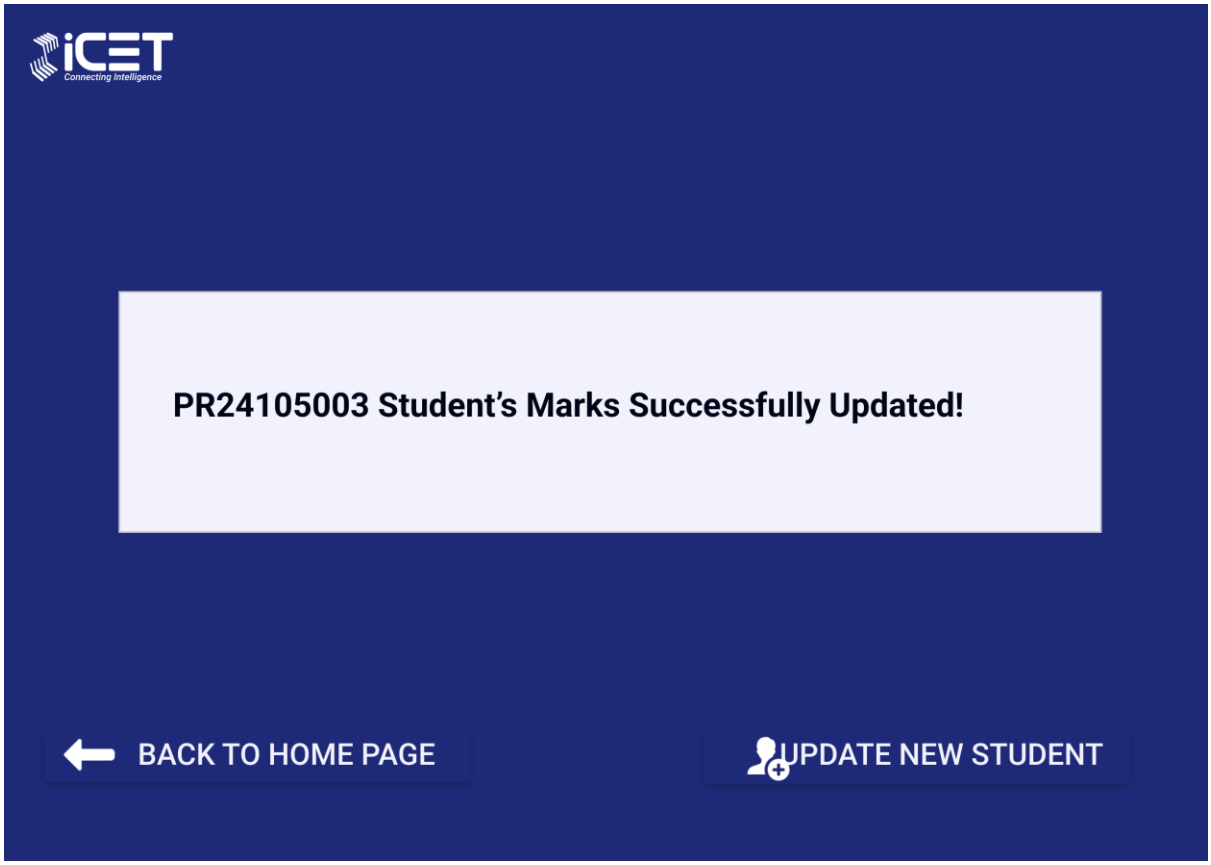


Figure 40 - Update already added mark successfully

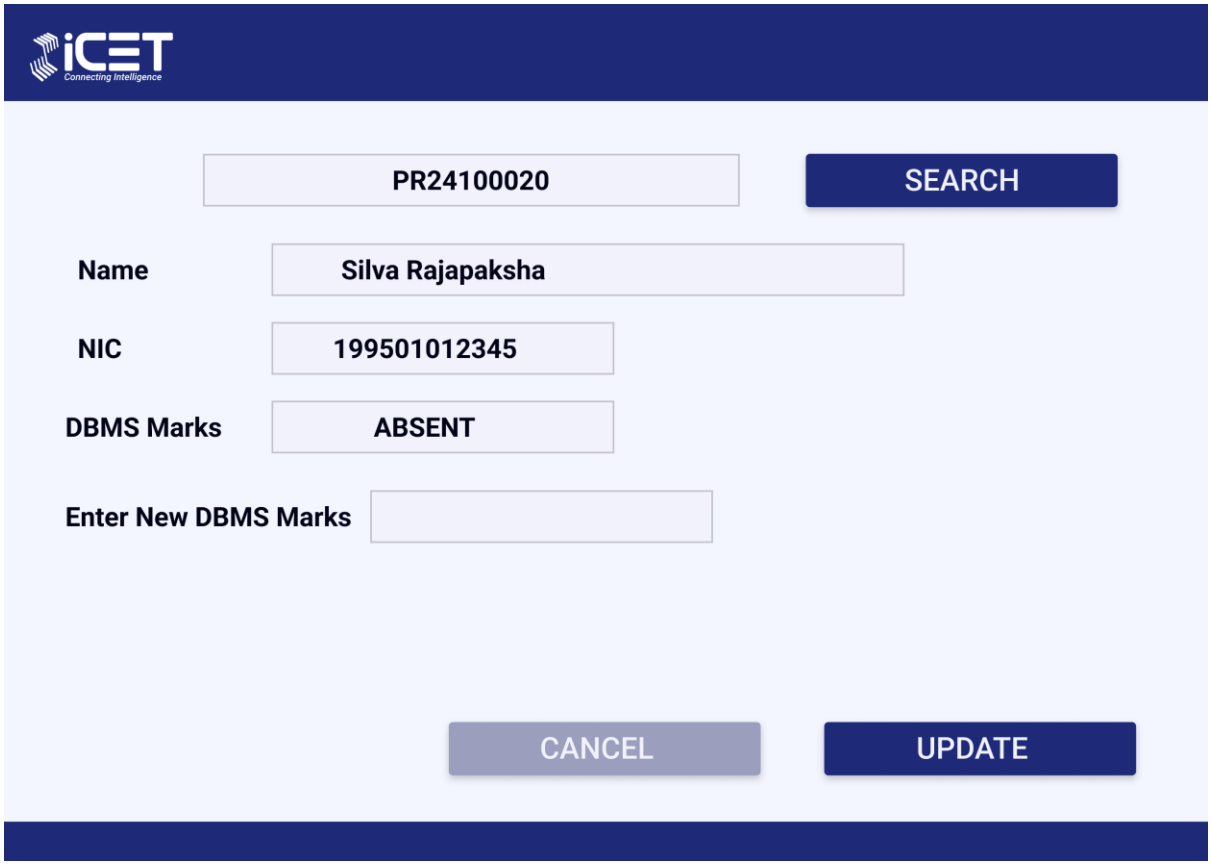


Figure 41 - Show current DBMS Marks

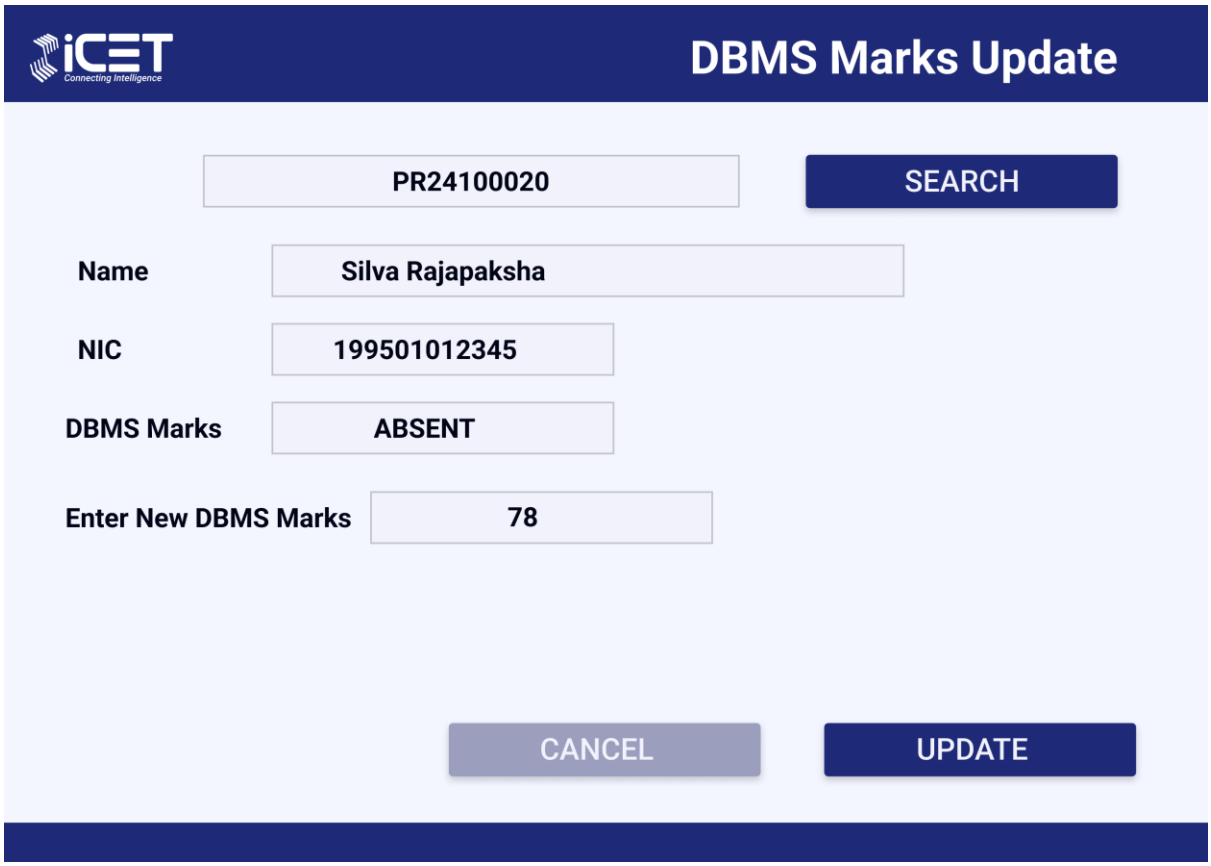


Figure 42 - Absent mark update

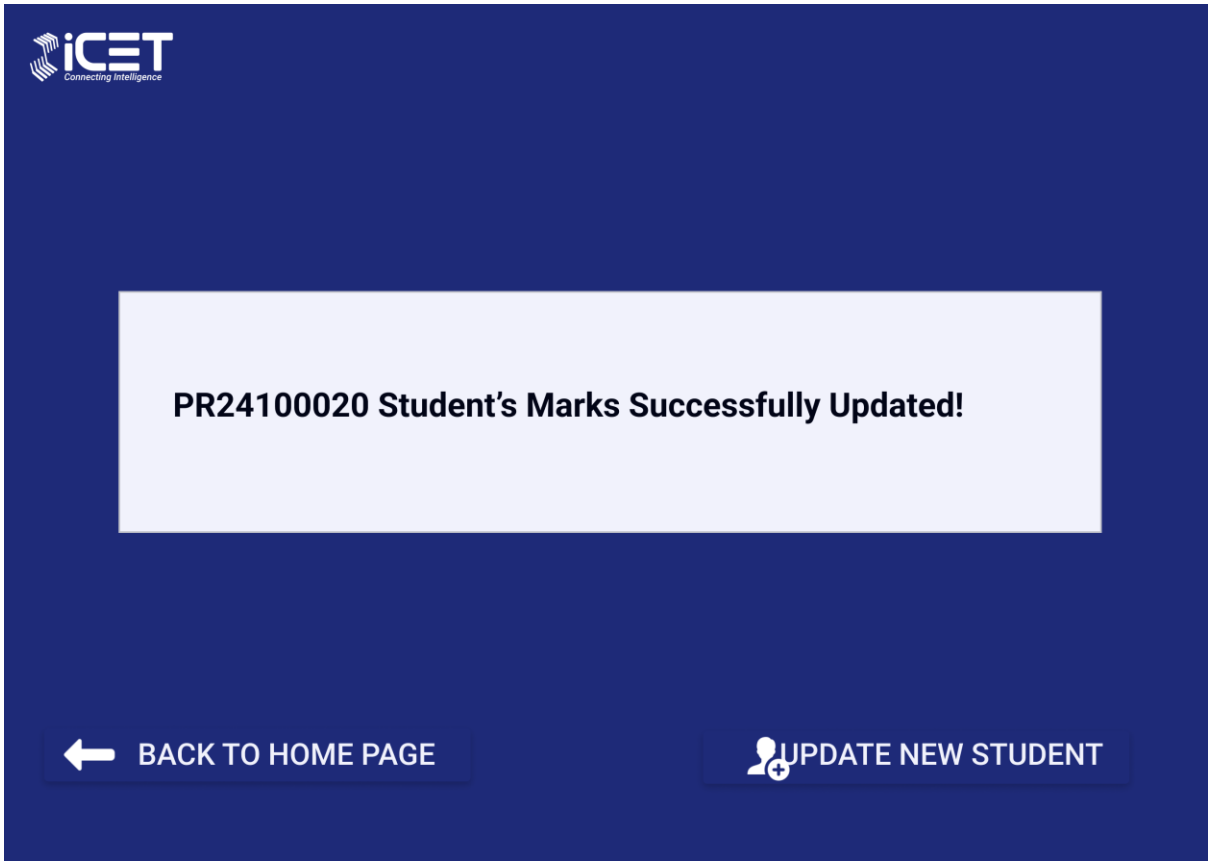


Figure 43 - Absent mark update successfully

[4] Report Management

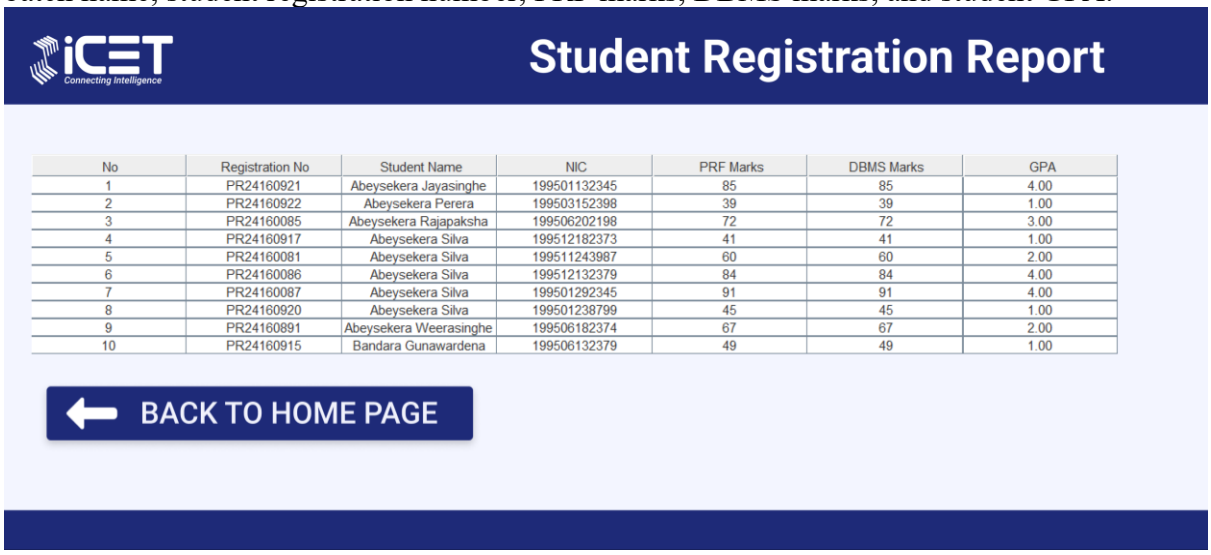
There are three options in Report Management, and the user can select the corresponding option they wish to execute.



Figure 44- Report Management Home Page

4.1 Student Registration Report

This displays all students with their details in alphabetical order by student name, including the batch name, student registration number, PRF marks, DBMS marks, and student GPA.



No	Registration No	Student Name	NIC	PRF Marks	DBMS Marks	GPA
1	PR24160921	Abeysekera Jayasinghe	199501132345	85	85	4.00
2	PR24160922	Abeysekera Perera	199503152398	39	39	1.00
3	PR24160085	Abeysekera Rajapaksha	199506202198	72	72	3.00
4	PR24160917	Abeysekera Silva	199512182373	41	41	1.00
5	PR24160081	Abeysekera Silva	199511243987	60	60	2.00
6	PR24160086	Abeysekera Silva	199512132379	84	84	4.00
7	PR24160087	Abeysekera Silva	199501292345	91	91	4.00
8	PR24160920	Abeysekera Silva	199501238799	45	45	1.00
9	PR24160891	Abeysekera Weerasinghe	199506182374	67	67	2.00
10	PR24160915	Bandara Gunawardena	199506132379	49	49	1.00

4.2 Batch-wise Student Report

First, the user should select the batch for which they want to view student data. After selection, the system will display all registered students in the chosen batch, including the batch name, student registration number, PRF marks, DBMS marks, and student GPA.

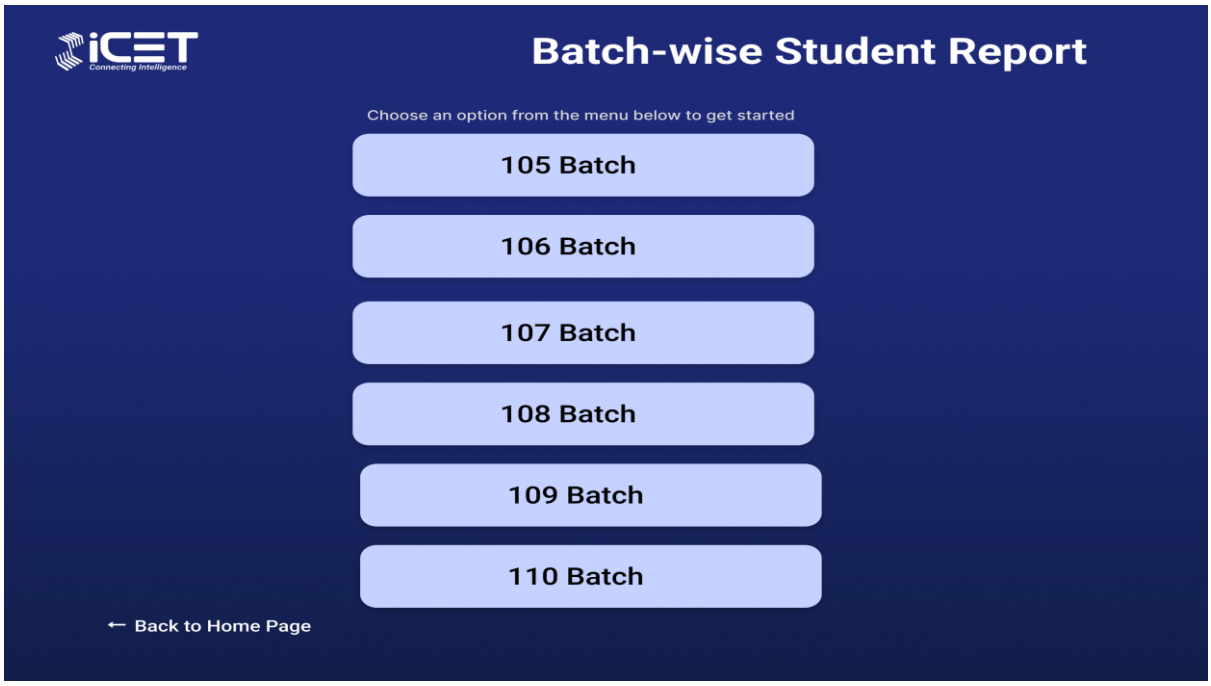
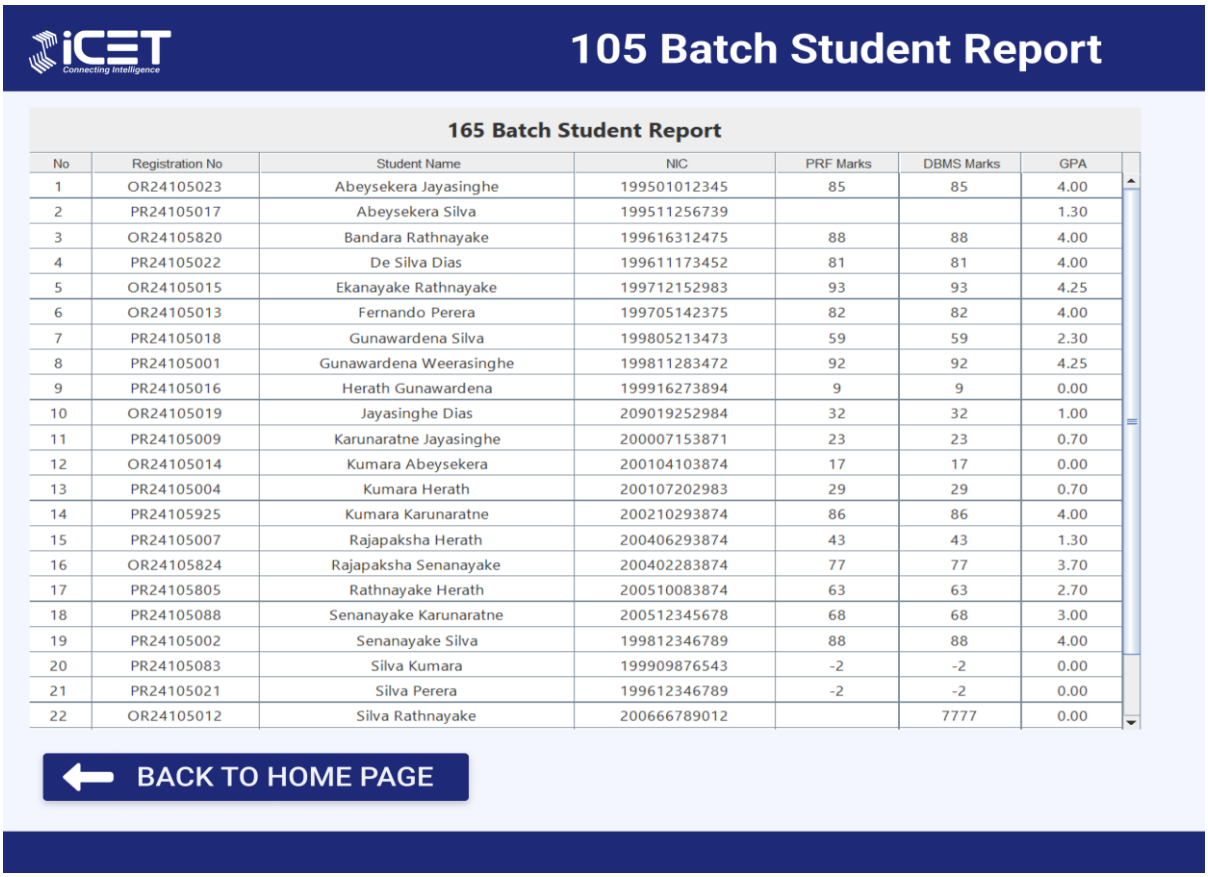


Figure 46- Batch Selection Page



No	Registration No	Student Name	NIC	PRF Marks	DBMS Marks	GPA
1	OR24105023	Abeysekera Jayasinghe	199501012345	85	85	4.00
2	PR24105017	Abeysekera Silva	199511256739			1.30
3	OR24105820	Bandara Rathnayake	199616312475	88	88	4.00
4	PR24105022	De Silva Dias	199611173452	81	81	4.00
5	OR24105015	Ekanayake Rathnayake	199712152983	93	93	4.25
6	OR24105013	Fernando Perera	199705142375	82	82	4.00
7	PR24105018	Gunawardena Silva	199805213473	59	59	2.30
8	PR24105001	Gunawardena Weerasinghe	199811283472	92	92	4.25
9	PR24105016	Herath Gunawardena	199916273894	9	9	0.00
10	OR24105019	Jayasinghe Dias	209019252984	32	32	1.00
11	PR24105009	Karunaratne Jayasinghe	200007153871	23	23	0.70
12	OR24105014	Kumara Abeysekera	200104103874	17	17	0.00
13	PR24105004	Kumara Herath	200107202983	29	29	0.70
14	PR24105925	Kumara Karunaratne	200210293874	86	86	4.00
15	PR24105007	Rajapaksha Herath	200406293874	43	43	1.30
16	OR24105824	Rajapaksha Senanayake	200402283874	77	77	3.70
17	PR24105805	Rathnayake Herath	200510083874	63	63	2.70
18	PR24105088	Senanayake Karunaratne	200512345678	68	68	3.00
19	PR24105002	Senanayake Silva	199812346789	88	88	4.00
20	PR24105083	Silva Kumara	199909876543	-2	-2	0.00
21	PR24105021	Silva Perera	199612346789	-2	-2	0.00
22	OR24105012	Silva Rathnayake	200666789012		7777	0.00

Figure 47- Batch Student Report

4.3 Industry Training Eligibility Report

In this option, the system will display all students who are eligible for Industry Training based on the selected criteria. The displayed details will include the batch name, student registration number, PRF marks, DBMS marks, and student GPA.

Industry Training Eligibility Criteria:

1. GPA must be greater than 3.25
2. Both PRF and DBMS marks must be greater than 50



No	Registration No	Student Name	NIC	PRF Marks	DBMS Marks	GPA
1	PR24195081	Gunawardena Weerasinghe	199581012345	85	66	3.50
2	PR24195807	Rajapaksha Herath	199502183764	60	30	3.35
3	PR24185809	Karunaratne Jayasinghe	199508153210	95	79	3.98
4	DR24105015	Ekanayake Rathnayake	199618312475	88	92	4.13
5	PR24165817	Abeysekera Silva	199683293481	73	84	3.65
6	OR24105023	Abeysekera Jayasinghe	199706253478	94	96	4.25
7	PR2410688	Kumara Gunawardena	199886193428	79	88	3.85
8	PR24106025	Rathnayake Kumara	199911983479	97	100	4.25
9	PR24187864	Silva Ekanayake	298868183471	88	74	3.65
10	PR24197806	Herath Abeysekera	208012043894	76	82	3.05
11	PR24187887	Rathnayake Fernando	208002193874	97	66	3.63
12	OR24107011	Abeysekera Silva	290101232984	90		4.13
13	DR24167813	Karunaratne Weerasinghe	290106273894	81	93	4.13
14	PR24107816	Weerasinghe Gunawardena	200112213874	67	5	3.50
15	PR24107921	Kumara Jayasinghe	299281913874	78	77	3.50
16	OR24107824	Kumara Weerasinghe	200200003471	60	100	
17	PR24198881	Fernando Rajapaksha	290212183471	86	79	3.85
18	PR24100803	Perera Wijesinghe	200204123894	91	87	4.13
19	OR24189812	Herath Perera	298302273874	88	83	4.98
20	OR24188814	Ekanayake Jayasinghe	290305123874		92	4.13
21	PR24188816	Rajapaksha Perera	200401153874	100	91	4.25
22	OR24188828	Kumara Herath	200418213874	90	58	3.63

Figure 48 - Batch Student Report